Survey: 6 Years of Neural Differential Cryptanalysis

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***Abstract*—At CRYPTO 2019, A. Gohr introduced Neural Differential Cryptanalysis and used deep learning to improve the state-of-the-art cryptanalysis of 11-round SPECK32. As of February 2025, according to Google Scholar, Gohr's article has been cited 229 times. The variety of targeted cryptographic primitives, techniques, settings, and evaluation methodologies that appear in these follow-up works grants a careful survey, which we provide in this paper. More specifically, we propose a taxonomy of these 229 publications and systematically review the 66 papers focusing on neural differential distinguishers, pointing out promising directions. We then highlight future challenges in the field, particularly the need for improved comparability of neural distinguishers and advancements in scaling. This survey helps researchers and engineers to identify the leading neural differential attacks, compare their performance, and highlight the outstanding open problems in AI-assisted cryptanalysis.**

***Index Terms*—Neural Differential Cryptanalysis, Survey**

Glossary -—Key Concepts for Non-Cryptographers

Block Cipher
A *symmetric-key* algorithm in which the same secret key is used for both encryption and decryption. It transforms an n-bit plaintext block into an n-bit ciphertext block through several rounds of non-linear substitution and linear permutation—effectively "scrambling" a bit sequence such as `01001...` into a key-dependent sequence like `01110...`. Formally, it is a keyed bijection

$$E_k : \{0,1\}^n \longrightarrow \{0,1\}^n.$$

PRP Security
The indistinguishability game in which an adversary receives black-box access to either the real cipher oracle $E_k(\cdot)$ or a uniform random permutation on n-bit blocks and must decide which oracle it is.

Differential Cryptanalysis
An analytical technique that encrypts input pairs differing by a chosen difference Δ, traces how Δ propagates through the rounds, and exploits the resulting output-difference statistics to infer information about the secret key.

Neural Differential Cryptanalysis
Differential cryptanalysis augmented with a neural network that learns to distinguish ciphertext data produced by the cipher from data produced by a random permutation.

Neural Distinguisher
A trained network that outputs "cipher" or "random" for a ciphertext (or ciphertext pair). Accuracy strictly better than 50% constitutes a successful distinguisher under the relevant security notion.

Neutral Bits (NBs)
Input-bit positions that can be flipped in both texts of a pair without breaking the targeted differential trail, yielding $2^{|NB|}$ additional usable pairs at negligible cost.

Wrong-Key Randomization
The working assumption that statistic(s) generated under an incorrect key follow the same distribution as those obtained from a truly random permutation, enabling efficient pruning of wrong key guesses.

I. Introduction

THE security of most digital applications relies on cryptography, the science of protecting the integrity, authenticity, and confidentiality of data. Confidentiality is about ensuring that only intended parties can read exchanged data. Typically, an *encryption key* is used in a secure *cipher* algorithm to encrypt the *plaintext* into a *ciphertext*. The recipient, knowing the *decryption key*, can easily retrieve the plaintext from this ciphertext. On the other hand, it is computationally intractable for an *adversary* who does not know the key.

The cornerstone of symmetric cryptography, where the encryption and decryption keys are the same, is *block ciphers*, which encrypt fixed-size messages, usually through iterations of a simple round function. Block ciphers play an important role in confidentiality but can also serve as building blocks to construct other primitives [1], such as hash functions and MAC schemes. Therefore, the security analysis of block ciphers (or cryptanalysis) is a crucially important field.

In the classical security notion, Pseudo Random Permutation (PRP) security, an adversary algorithm is assumed to have black-box access to an oracle function, implementing either (A) the studied block cipher (with a hidden, random key) or (B) a random permutation. A block cipher is considered secure under this notion if no such adversary can distinguish between situation A and B faster than using the trivial strategy of enumerating all possible keys to find one that matches the oracle's output (A) or be convinced that no such key exists (B). On the other hand, if a *distinguisher* exists, the block cipher is considered broken, as a good distinguisher can usually be used to retrieve the key. The performance of a distinguisher is usually expressed in terms of *time complexity* (number of operations to be performed by the attacker), *data complexity* (number of queries to the oracle), and *memory complexity*.

The main goal of cryptanalysis is to estimate how many iterations of the round functions (or *rounds*) are needed for security. This is an iterative process, and new results continue to be published regularly years after the release of a cipher. Therefore, cryptographers are eager to build and improve tools that help with this tedious task. Differential cryptanalysis, first introduced by Biham and Shamir in 1991 [2], examines how input differences (δ) propagate through block ciphers, seeking high-probability differentials where specific plaintext differences yield predictable ciphertext differences (Δ). Tradi-

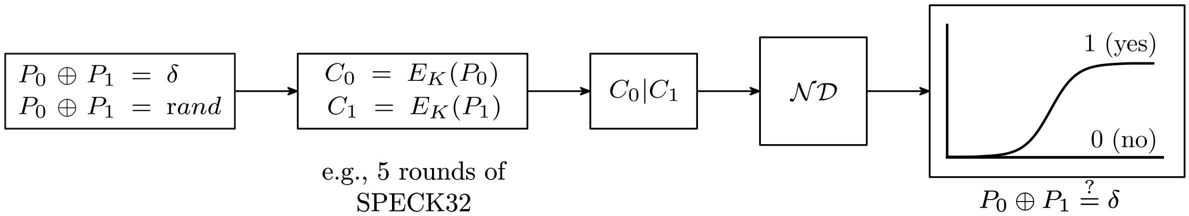

Fig. 1. **Neural Differential Distinguisher: Basic Training Pipeline.** Start with two plaintext P_0, P_1, where $P_0 \oplus P_1 = \delta$ or $P_0 \oplus P_1 = \text{rand}$. Encrypt them using a symmetric key K to obtain ciphertexts C_0, C_1. Concatenate the ciphertexts $C_0|C_1$ and input them into a neural distinguisher $\mathcal{ND}$. The neural distinguisher's output is a neuron with a sigmoid activation function. The sigmoid curve indicates a binary decision output to answer if $P_0 \oplus P_1 = \delta$.

tional approaches rely on mathematical analysis of difference propagation through cipher rounds, calculating probabilities that given input differences produce specific output differences to create distinguishers. While this technique proved devastatingly effective against early ciphers, modern designs incorporate specific countermeasures that make conventional differential analysis increasingly challenging and computationally intensive, requiring extensive manual analysis of cipher structure and often yielding suboptimal distinguishers for complex round functions. Deep learning, due to its strength at detecting and distinguishing patterns, has long been seen as a potential candidate to assist the task of cryptanalysis.

Deep learning has experienced significant advancements in recent years, leading to remarkable achievements in various domains. Initially, Frank Rosenblatt introduced Multi-Layer Perceptrons (MLPs) in his 1958 book Perceptron and laid the foundation for modern neural networks. The introduction of Convolutional Neural Networks (CNNs) in the 1980s [3] led to a breakthrough in computer vision in the form of LeNet, which achieved human-level performance in digit recognition in 1998 [4]. Through advancements in Monte Carlo Tree Search (MCTS) and reinforcement learning, further leaps were enabled, such as Google's AlphaGo surpassing human capabilities [5], [6], [7]. More recently, transformer-based Large Language Models (LLMs) [8], such as GPT, have revolutionized natural language processing, demonstrating near-human capabilities in tasks like machine translation and language generation.

Despite the long-standing recognition of the intersection between cryptography and machine learning [9], [10], [11], the use of computational intelligence in cryptanalytic tasks has remained limited. Early approaches typically relied on extensive precomputation [12], exploited implementation flaws (e.g., side-channel attacks) [13], targeted inherently weak cryptographic schemes [14], [15], or generally proved ineffective [16]. It was not until Gohr's seminal work [17], presented at CRYPTO 2019, that a breakthrough was achieved by combining deep learning with traditional cryptanalytic techniques. Gohr's work was the first to demonstrate that neural networks could be successfully leveraged in cryptanalysis, producing attacks that improved upon state-of-the-art techniques against a round-reduced version of a modern block cipher.

Gohr's groundbreaking work [17] established a new paradigm in cryptanalysis by demonstrating that deep neural networks could serve as superior distinguishers in what is now termed *neural differential cryptanalysis*. Despite their black-box nature, these networks consistently outperformed traditional methods by learning to identify subtle patterns in ciphertext pairs that indicate whether they originated from structured versus random input differences, as illustrated in Figure 1. Gohr's approach proved particularly effective against the NSA-designed SPECK32 [18] cipher, demonstrating superior accuracy on 8-round variants while significantly reducing time complexity for 11-round key recovery attacks. Once trained, these neural distinguishers serve as critical components in practical recovery attacks, as illustrated in Figure 2, fundamentally challenging conventional cryptanalysis assumptions and enabling automated analysis of cipher variants without extensive manual mathematical investigation.

Since Gohr's seminal paper, researchers have explored every part of the basic Neural Differential Cryptanalysis pipeline. A majority of the works citing Gohr that focus on neural differential distinguishers have attempted to apply the scheme to **24 different symmetric-key primitives** (with SPECK and SIMON families receiving the most attention at 38 and 36 experiments respectively), improve the distinguishing advantage by **developing 15+ distinct network architectures** (from the dominant $\mathcal{ND}_{\text{Gohr}}$ to specialized approaches like DBitNet, SE-ResNet, and even quantum neural networks), **scaling training data from 20K to 32.48 billion samples** (though 20M remains the de facto standard), or **innovating sample formats** (including advanced feature engineering techniques that have been shown to improve distinguishing accuracy). Finally, we also observe emerging research directions aimed at enhancing neural distinguishers across multiple dimensions: increasing **automation** of the attack pipeline (exemplified by DBitNet's cipher-agnostic approach and AutoND flags in 21 experiments), improving **transparency** through explainability techniques that reveal the cryptographic features being learned (as seen in ensemble approaches), and boosting **effectiveness** of the obtained neural distinguishers in practical **key recovery** attacks.

The explosive growth of neural cryptanalysis literature has created an increasingly fragmented landscape that poses significant barriers to systematic knowledge synthesis. This rapid expansion has outpaced efforts to establish coherent theoretical foundations and standardized methodologies, creating particular challenges for interdisciplinary researchers who must

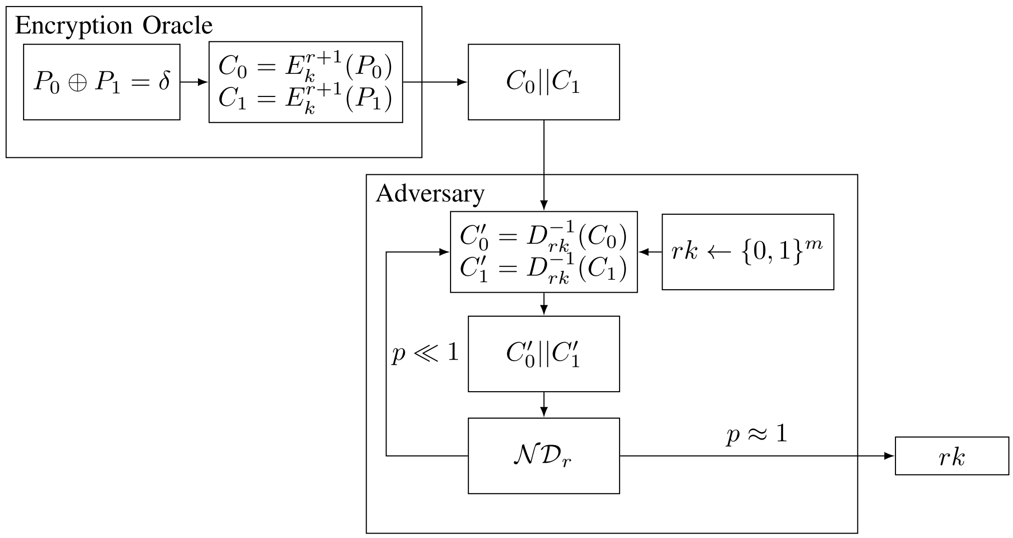

Fig. 2. **Neural Differential Distinguisher: Basic Key Recovery Pipeline.** An encryption oracle generates an $(r+1)$-round ciphertext pair with input difference δ. The adversary decrypts one round using a key guess rk and feeds the result to a neural distinguisher trained on the r-round distribution. Keys producing prediction scores near 1 are output as candidates, following the wrong-key randomization hypothesis [17].

navigate both cryptographic principles and machine learning techniques. The absence of systematic knowledge organization has fostered several critical problems: redundant research efforts where teams independently investigate nearly identical questions while remaining unaware of parallel work, leading to duplicated findings and contradictory claims such as [19]'s erroneous assertion of developing the first truncated neural distinguisher; fundamental disagreements on core methodological questions, exemplified by the conflicting assessments of neural architecture suitability among [20], [21], and [22]; and persistent misinterpretations of foundational concepts, particularly regarding the proper aggregation of multiple ciphertext pair predictions – an issue that Gohr explicitly addressed in [23] yet continues to be misunderstood in subsequent work. As this field's rapid expansion shows no signs of abating, these structural problems will inevitably intensify without intervention, underscoring the urgent need for a comprehensive survey that can establish coherent theoretical frameworks and standardized practices in Neural Differential Cryptanalysis.

a) Our Contributions.: In our survey, we have achieved the following:

1) **Comprehensive Field Review:** We conducted an exhaustive survey of the follow-up work (section V). In this process, we have identified the **full body of research** in the field of Neural Differential Cryptanalysis. We analyzed the directions of the field, resulting in a detailed taxonomy of Neural Differential Cryptanalysis (section V).
2) **Explainability and Key Recovery Overview:** We provide a comprehensive overview of recent advancements in explainability techniques using neural differential distinguishers (section VI). Since analyzing neural distinguishers constitutes the core contribution of our work, we provide a comprehensive overview of advancements in neural-aided key recovery (section VII) as a contextual application of our findings.
3) **Rigorous Classification and Comparison:** We systematically classify and compare peer-reviewed research outcomes on neural differential distinguishers (section VIII), across various techniques, architectures, and primitives. We also identify promising research directions and severe methodological issues in some peer-reviewed papers and challenge their results.
4) **Best Practice Recommendations:** Evaluating research involving the training of neural networks presents significant challenges. We have developed a comprehensive set of best-practice guidelines specifically tailored for reviewers of Neural Differential Cryptanalysis research (section IX).
5) **Future Challenges:** Finally, we identify and discuss two major challenges set to shape the next six years of neural cryptanalysis (section X).

II. Related Work

While a substantial body of literature has focused on developing and analyzing effective neural differential distinguishers, this paper is, to the best of our knowledge, the first to systematically organize a large collection of research (229 papers) and highlight promising directions and challenges in this area. Recent surveys [24], [25], [26], [27], [28] do not claim a systematic approach, cover a significantly smaller body of work, and most lack a specific focus on machine learning-based cryptanalysis.

Bellini *et al.* in [24] examine machine learning-based black-box and white-box cryptanalysis. Regarding white-box cryptanalysis, they reference Gohr's work [17] along with 15 related follow-up studies, though these were not selected systematically and include preprints. Nitaj and Richidi, in [25], explore various cryptographic areas that could benefit from the application of artificial intelligence (AI). While they briefly mention the potential of machine learning to enhance side-channel and cryptanalytic attacks on symmetric block ciphers, they neither provide a systematic analysis of existing work in this area nor delve into the specific methodologies involved. Awad and El-Alfy, in [29], conduct a survey on computational intelligence applications in cryptography, with a focus on the automated design and cryptanalysis of ciphers. However, their work predates Gohr's introduction of differential machine learning-based cryptanalysis in [17], and as a result, it does not include a comprehensive review of neural distinguishers found in more recent literature. Singh *et al.* [28] investigate various machine learning and optimization techniques, including Hill Climbing and Particle Swarm Optimization, applied to cryptanalysis. They also reference Gohr's research [17] along with 12 subsequent studies that build upon it. However, the selection of these papers is not based on a systematic methodology. The work by Martinez *et al.* [27] is the most comparable to ours. Although it does not follow a systematic paper selection process, it aims to capture the state-of-the-art and categorizes 10 works, including Gohr's, based on their architectures and the cryptographic schemes they target.

III. AI and Cryptography in the Beginnings

The popularity and widespread adoption of neural differential distinguishers (more precisely, deep learning-based cryptanalysis) can be credited to the seminal work of A. Gohr [17]. However, even in that work, the author mentioned a number of related works at the intersection between cryptanalysis and AI. What distinguishes Gohr's work from previous ones is that it considers relevant (modern) ciphers and manages to obtain results that surpass state-of-the-art conventional cryptanalysis techniques. The following section is not meant to provide an exhaustive list of works connecting AI and cryptology but rather provide a brief historical overview of various approaches.

Already in 1947, researchers started considering connections between cryptography and artificial intelligence [9]. While this attempt was devoid of any technical details, it still showcases the interest of the scientific community in combining these two domains. In 1984, L. Valiant discussed learnable Boolean functions and mentioned the evidence from cryptography that the whole class of functions computable by polynomial-size circuits is not learnable [10]. Shortly after, in 1988, Minsky and Papert showed that every Boolean function can be realized by an MLP neural network [30]. In 1994, R. Rivest wrote a paper on connections between cryptography and machine learning [11]. Already there, he mentioned the possibility of using machine learning for cryptanalysis.

In 2002, Klimov *et al.* analyzed the security of a key exchange protocol based on mutually learning neural networks [31]. While the authors experimentally verify that it is unlikely for a particular attacker using a similar neural network to converge to the same key, they still break the protocol using more advanced cryptanalytic techniques. Similarly, in 2016, Abadi and Andersen employed neural networks in a framework inspired by generative adversarial networks (GANs) to develop an encryption scheme [32]. Although this early research did not show any formal security, Coutinho *et al.* demonstrated in 2018 [33] that, with certain architectural modifications, the network could be trained to learn the One-Time Pad [34].

In 2002, Castro *et al.* used evolutionary algorithms to construct a cryptanalytic tool that can distinguish between the two-round TEA algorithm and random permutations [35]. In 2007, Laskari *et al.* considered the application of diverse computational intelligence techniques to the cryptanalysis of known cryptosystems, including public key cryptosystems and Feistel ciphers [36]. In the same year, Tapiador *et al.* used heuristics to conduct nonlinear cryptanalysis and applied it to the MARS cipher S-box [37]. In 2012, Chou *et al.* experimented with machine learning techniques to mount distinguishing attacks and concluded it is not possible to extract useful information from ciphertexts produced by modern ciphers operating in secure modes, nor to distinguish them from random data [26]. On the other hand, Svenda *et al.* in 2014 used evolutionary algorithms to construct empirical tests for randomness [38].

Finally, in 2017, Awad and El-Alfy surveyed computational intelligence applications in cryptography, focusing on the automated design and cryptanalysis of ciphers [29].

IV. Preliminaries

This section introduces key concepts in machine learning-assisted differential cryptanalysis: conventional differential cryptanalysis (subsection IV-A), deep learning applications (subsection IV-B), and neural network-aided key recovery (subsection IV-C).

A. Differential Cryptanalysis

Differential cryptanalysis [2] is a chosen plaintext attack analyzing how plaintext perturbations propagate through ciphers. While typically using bitwise XOR differences, some approaches employ modular addition or rotations. For a map $F : \{0,1\}^b \rightarrow \{0,1\}^b$, a *differential transition* is a pair $(\delta, \Delta) \in \{0,1\}^b \times \{0,1\}^b$ with probability:

$$P(\delta \rightarrow \Delta) = \frac{\left|(\{x \in \{0,1\}^b : F(x) \oplus F(x \oplus \delta) = \Delta\})\right|}{2^b}.$$

B. Training Neural Differential Distinguishers

For plaintexts $p_1, p_2 \in \{0,1\}^b$ with ciphertexts $c_i = F(p_i) \in \{0,1\}^b$, neural distinguishers approximate the function for fixed difference $\delta \in \{0,1\}^b$:

$$Y(c_1||c_2) = \begin{cases} 1, & \text{if } p_1 \oplus p_2 = \delta, \\ 0, & \text{else.} \end{cases}$$

Success requires identifying nonrandom properties in output distributions resulting from input difference δ. Training typically uses balanced datasets: 50% samples $(c_1, c_2, 0)$ with random p_1, p_2, and 50% samples $(c_1, c_2, 1)$ where $p_2 = p_1 \oplus \delta$. Networks are trained via stochastic gradient descent [39][1] using loss functions such as mean squared error.

[1] We introduce essential machine learning terminology needed to understand the techniques used in neural differential cryptanalysis: Stochastic gradient descent is an iterative optimization method that updates the weights of a neural network by calculating error gradients on small random subsets ("batches") of the training data rather than the entire dataset. The "loss function" (e.g., mean squared error) quantifies prediction error, while an "epoch" represents one complete pass through the training dataset. The Adam optimizer is an advanced gradient descent variant that adapts learning rates individually for each weight. L2 regularization prevents overfitting by penalizing large weight values, essentially constraining the model's complexity.

Following Gohr [17], effective implementations use approximately 10^7 training samples, 10^6 testing samples, batch sizes around 5000, and up to 200 training epochs. Performance enhancements often include Adam optimizer [40], L2 regularization [41], and cipher-specific architectures [23], [42].

C. Neural-aided Key Recovery

Neural distinguishers $\mathcal{ND}$ that approximate $Y(c_1||c_2)$ enable practical key recovery attacks on block ciphers, demonstrating their concrete cryptanalytic value. This section outlines the attack methodology based on Gohr's seminal approach [17], which has become the standard framework in subsequent research. We denote the r-round reduced block cipher with secret key K as F_K^r.

a) The Basic Attack: The attack leverages a pre-trained neural distinguisher ND_r for F^r to compromise F_K^{r+1} with secret key K. For example, a 5-round SPECK32/64 distinguisher enables attacks on 6-round SPECK32/64.

The attack targets the last round key k_{r+1} in round-based ciphers that use function f_k with keys $k_1, \ldots, k_{r+1}$ derived from master key K and begins by querying the oracle F_K^{r+1} with a conforming pair p_1 and $p_2 = p_1 \oplus \delta$, obtaining ciphertext pair (c_1, c_2). Next, for some random key guess k', the attacker computes $c_i' = f_{k'}^{-1}(c_i)$ and evaluates $R = D_r(c_1', c_2')$. We rank key candidates by prediction score, as the correct key yields $R \approx 1$ (the distribution matches what ND_r was trained to recognize), while incorrect keys produce $R < 1$, following the wrong-key randomization hypothesis [17].

For SPECK32, where round keys (16 bits) are smaller than the master key (64 bits), we can feasibly enumerate all candidates. After identifying the last round key, the process can be repeated to recover earlier round keys until the entire sequence is reconstructed.

Our simplified attack explanation omitted that prediction scores exhibit variance, which can be mitigated by aggregating scores across multiple ciphertext pairs for each key candidate, thereby enhancing the statistical reliability of the distinguisher. In [17], the responses for a given key guess k' are aggregated into a single score by the equation:

$$s_{k'} = \sum_{i=1}^{n} \log_2 \left(\frac{R_i^{k'}}{1 - R_i^{k'}} \right),$$

where R_i^k represents the distinguisher's response for the i-th ciphertext pair.

b) Extending the Rounds Covered by the Distinguisher.: Neural distinguishers can effectively extend their coverage by exploiting structural properties of ARX ciphers. For ciphers such as Speck and Simon, where initial subkey addition occurs after the first nonlinear operation, neural distinguishers gain an additional round without computational overhead. This extension is achieved by constructing plaintext pairs that deterministically yield ciphertext differences corresponding to the neural distinguisher's trained input difference δ following the first round transformation.

Furthermore, neural distinguishers can be combined with classical differential cryptanalysis through a hybrid approach that prepends an s-round classical differential transition to an r-round neural distinguisher, yielding an $(s + r)$-round distinguisher. This combination introduces a fundamental complexity tradeoff inherent to differential cryptanalysis. In Gohr's implementation [17], the selected 2-round differential transition exhibits probability $\frac{1}{64}$, necessitating an average increase of 64-fold in data complexity while extending the 7-round neural distinguisher to achieve 9-round coverage.

The integration of classical and neural techniques presents a significant challenge in score aggregation. When prepending classical differentials, the aggregation of scores across multiple ciphertext pairs becomes problematic due to non-conforming pairs that introduce stochastic noise, degrading distinguisher performance. This limitation is addressed through the strategic deployment of *(Probabilistic) Neutral Bits* (PNBs), which enhance the signal-to-noise ratio. For a neutral bit i, pairs (p_1, p_2) satisfying differential $\delta \to \Delta$ ensure that $(p_1 \oplus (1 \ll i), p_2 \oplus (1 \ll i))$ also conform to the differential with probability 1 (or high probability for PNBs). Consequently, utilizing j PNBs enables the construction of plaintext structures containing 2^j pairs where conformance to the prepended differential exhibits binary behavior: either all pairs within the structure satisfy the differential or none do, thereby ensuring consistent scoring mechanisms. This methodology was subsequently extended to incorporate conditional simultaneous neutral bit-sets and switching bits for adjacent differentials [43].

c) Reducing Computational Cost.: Gohr's approach achieves significant computational complexity reduction through the application of Bayesian optimization to key search procedures [17]. This methodology represents a departure from exhaustive key enumeration toward statistically informed search strategies. The technique initiates by constructing a *Wrong Key Response Profile* (WKRP) that characterizes the distributional properties of neural distinguisher responses under incorrect key hypotheses.

For a ciphertext pair (c_1, c_2) derived from plaintexts p and $p \oplus \delta$, with correct r-round key k and candidate key $k' = k \oplus \gamma$, the distinguisher response is modeled as:

$$R_{D,\gamma}(c_1, c_2) = D\left(f_{k'}^{-1}(c_1), f_{k'}^{-1}(c_2)\right)$$

where D represents the neural distinguisher and $f_{k'}^{-1}$ denotes single-round decryption. The response variable $R_{D,\gamma}$ is characterized by a normal distribution with parameters μ_γ and σ_γ that depend functionally on the key difference γ.

The key search algorithm operates iteratively, employing an acquisition function derived from the WKRP to optimize candidate selection. Given observed responses $R_1, \ldots, R_n$ corresponding to key candidates $k_1', \ldots, k_n'$, the optimization procedure selects subsequent candidates k by minimizing:

$$\sum_{i=0}^{n-1} \frac{(R_i - \mu_{k \oplus k_i'})^2}{\sigma_{k \oplus k_i}^2}$$

This formulation aligns the precomputed wrong key response profile optimally with empirically observed values, typically converging within a limited number of iterations compared to exhaustive round key enumeration.

To mitigate computational overhead from unpromising search directions, Upper Confidence Bounds (UCB) serve as algorithmic stopping criteria. For t independent encryption oracles F_K^{r+1}, the attack prioritizes instances according to:

$$sk = w_{\max}^i + \alpha \cdot \sqrt{\frac{\log_2(j)}{n_i}}$$

where $w_{\max}^i$ represents the maximum distinguisher score achieved for instance i, n_i denotes the computational iterations allocated to instance i, j corresponds to the current global iteration, and $\alpha = 10$. This formulation implements an exploration-exploitation tradeoff that concentrates computational resources on instances demonstrating either insufficient exploration or elevated key candidate scores.

d) Additional Verification.: The verification phase addresses the challenge of near-correct key candidates producing high neural distinguisher scores. Candidates differing by 1-2 bits from the correct key often generate elevated scores, requiring localized search within the Hamming neighborhood of promising candidates to identify the correct key with marginally superior scores.

Gohr implements joint recovery of $(r+1)$-round and r-round keys by combining distinguishers D_r and D_{r-1}. The protocol triggers r-round key search using D_{r-1} when an $(r+1)$-round candidate exceeds threshold t_1, returning both keys only when an r-round candidate surpasses threshold t_2. This cascaded approach provides verification, as incorrect $(r+1)$-round keys rarely generate elevated scores for corresponding r-round hypotheses, reducing false positives.

V. Neural Differential Cryptanalysis: A Taxonomy of Research Directions

A. Selected Literature

As of February 03, 2025, a total of 229 works cite Gohr's work [17] on Google Scholar. Among these, we discarded 4 references that were either redundant or not linked to a paper, and 33 that were not available in English. Additionally, 34 references are not peer-reviewed, and only available as preprints [44], [45], [46], [43], [47], [48], [49], [23], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75]. After excluding these, we are left with 158 peer-reviewed references, which we systematically categorize as shown in Figure 3.

We consider the following references outside the field of research on Neural Differential Cryptanalysis as they are surveys, overviews, theses, or book chapters that treat the use of "ML in cryptography" [24], [76], [77], [27], [78], [25], [79], [80], [81], [82], [83], [84], [85], or their research focuses on other topics such as: classical cryptanalysis [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], the theory of Neural Differential Cryptanalysis [98], cryptanalysis of historic or toy ciphers [99], [100], [101], [102], [103], [104], deep learning-supported design of cryptographic algorithms [105], [106], [107], [108], [109], [110], [111], neural side-channel attacks [112], [113], [114], [115], distinguishers between different ciphers [116], [117], [118], [119], neural preimage attacks [120], [121], [122], the introduction of a new tool or library [123], [124], [125], [126], [127], [128], the proposal of a new cipher [129], [130], [131], [132], neural output prediction attacks [133], [134], [135], [136], neural integral distinguishers [137], [138], [139], neural attacks on protocols [140], [141], post-quantum schemes [142], [143], pseudorandom number generators [144], [145], or other unrelated topics [146], [147], [148], [149], [150], [151], [152], [153], [154], [155], [156], [157], [158], [159], [160], [161], [162], [163], [164], which leaves us with a total of 66 peer-reviewed publications in the field of Neural Differential Cryptanalysis.

The Body of Peer-Reviewed Research in Neural Differential Cryptanalysis

The full body of peer-reviewed publications that focus specifically on advancing research of Neural Differential Cryptanalysis is [20], [165], [22], [166], [167], [168], [169], [42], [170], [171], [172], [173], [174], [175], [176], [177], [178], [179], [180], [181], [182], [183], [184], [185], [186], [187], [188], [189], [190], [191], [192], [193], [194], [195], [196], [197], [198], [199], [200], [201], [21], [202], [203], [204], [205], [19], [206], [207], [208], [209], [210], [211], [212], [213], [214], [215], [216], [217], [218], [219], [220], [221], [222], [223], [224], [225]

B. Taxonomy of Research Directions

We found contributions to the explainability (or interpretability) of neural distinguishers in the following 17 works [165], [167], [169], [173], [175], [176], [17], [178], [180], [182], [190], [193], [194], [203], [19], [217], [225], and will discuss their respective contributions in section VI.

We found contributions to neural-aided key recovery attacks in the following 22 works [166], [169], [173], [174], [17], [181], [182], [187], [186], [189], [193], [194], [201], [19], [206], [207], [212], [217], [224], [222], [219], [225], and will give an overview of these works in section VII.

Most (62/66) of peer-reviewed research on Neural Differential Cryptanalysis involves training neural differential distinguishers. More precisely, neural differential distinguishers are trained in [165], [20], [22], [166], [167], [42], [168], [169], [171], [172], [173], [174], [175], [176], [177], [179], [181], [180], [182], [183], [184], [185], [187], [186], [188], [190], [191], [189], [192], [193], [194], [195], [197], [199], [200], [201], [21], [202], [204], [203], [19], [205], [206], [207], [208], [209], [210], [211], [212], [213], [214], [215], [216], [217], [218], [220], [221], [222], [224], [223], [219], [225].

A comparative review of the peer-reviewed neural differential distinguishers from 55 papers (including Gohrs' seminal work [17]) is provided in subsection VIII-C. We excluded papers that were inaccessible [170], [198], focused primarily on explainability [203], lacked concrete accuracy measurements [181], utilized leakage models outside conventional security assumptions [205], [183], or prioritized input difference compatibility over distinguisher performance in hybrid approaches [216], [213].

Two recent papers [182], [193] investigating neural differential attacks on large-state block ciphers predominantly

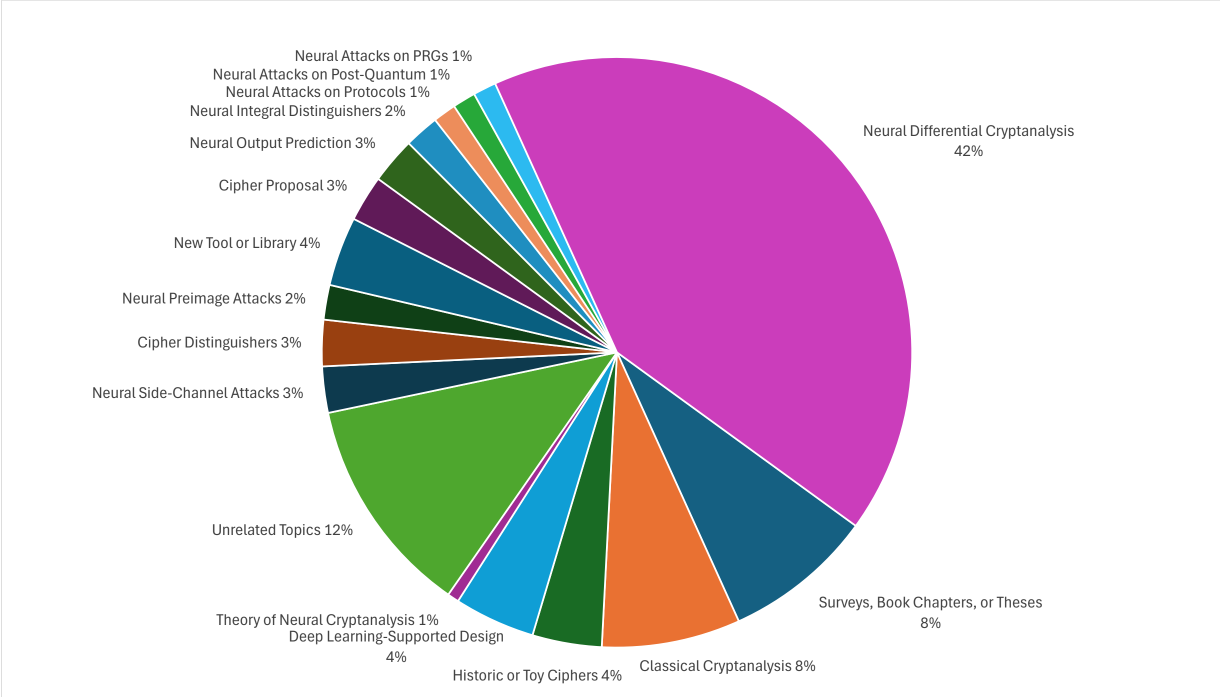

Fig. 3. Our taxonomy of the peer-reviewed English-language publications that cite Gohr's seminal work [17].

emphasize key recovery methodologies while providing limited insights into their neural network training processes. This methodological opacity presents significant challenges for our comparative review. However, Huang *et al.* [182] provide their implementation, enabling a more comprehensive evaluation of their approach despite the initial presentation's technical brevity.

Gohr's analysis was performed within the secret key chosen-plaintext attack (SK/CPA) model. We do not consider the work of Phan *et al.* [196] as it operates under a fundamentally different adversary model, where generative AI techniques are trained in an adaptively chosen ciphertext or known key scenario to distinguish 10-round SPECK32/64, making direct comparison inappropriate.

VI. Overview: Neural Differential Distinguisher Explainability

Neural distinguishers (neural networks that can identify non-random patterns in encrypted data), enabling new cryptanalytic attacks, potentially better than manual cryptanalysis, motivated researchers to try to understand what made these attacks so powerful and to learn new properties from them. The lack of explainability is the "machine's inability to explain its decisions and actions to human users" [226]. One of the major efforts in research on explainability was the 4-year program (2017-2021)" XAI" by the Defense Advanced Research Projects Agency (DARPA) of the United States Department of Defense *"DARPA's Explainable Artificial Intelligence (XAI) Program"* [227]. A more recent review of the research in XAI is given in *"Interpreting Black-Box Models: A Review on Explainable Artificial Intelligence"* [228]. To this day, explainability is an active research field in AI and has resulted in various ways to add some explainability to a neural network, e.g., by pruning, ablation studies, or visualization techniques.

a) Understanding the learned cryptanalytic features: A. Gohr investigated the capabilities of provided neural networks by introducing a differential cryptanalytic task called the real differences experiment [17]. This experiment involved applying a uniform random mask to both ciphertexts in a candidate pair, thereby preserving the differential relationship (the XOR difference pattern between encrypted pairs) while mathematically obscuring the actual bitwise difference from the neural network. Statistical analysis of the neural distinguisher's performance demonstrated that it maintained classification accuracy exceeding that of random guessing. This empirical evidence suggests that such neural architectures exploit cryptographically relevant features beyond those captured in conventional difference distribution tables (DDTs, which are lookup tables storing how input differences propagate through a cipher).

In [165], Benamira *et al.* studied the properties of pairs that were correctly classified and proposed that Gohr's neural distinguishers learn differential-linear features (a combination of differential cryptanalysis, which tracks difference propagation, and linear cryptanalysis, which exploits linear approximations). In particular, the authors observed that the pairs for which the score of the neural distinguisher at round 5 is high often follow a specific truncated differential pattern (a partial difference pattern affecting only some bits) at round 3; a similar observation is made for rounds 6 and 4, leading to the authors proposing that the features learned by the neural distinguisher are differential-linear in nature. Later, the truncated differential observations from [165] were used by [42] to identify good input differences for neural distinguishers automatically.

The authors further modified the convolutional neural network to use a Heaviside activation function in the first layer, which binarizes the associated intermediate values, allowing them to analyze the linear transformation that the network learned for SPECK. Their findings demonstrate that this first convolutional layer primarily extracts sophisticated features following the pattern $(C_1, C_2) = (l_1||r_1, l_2||r_2) \rightarrow (l_1 \oplus l_2, l_1 \oplus r_1 \oplus l_2 \oplus r_2, l_1 \oplus r_1, l_2 \oplus r_2)$, where l and r represent left and right halves of ciphertext blocks. Further experimentation revealed that replacing Gohr's network's initial 1D convolutions with a static feature extractor maintained comparable accuracy levels, suggesting the importance of these specific transformations. The researchers further revealed that subsequent convolutional layers effectively learn to encode a compressed (generalized) representation of the differential distribution table. Additionally, they demonstrated that the prediction head could be replaced with various alternative ensemble classifiers without significant degradation in performance, maintaining nearly identical accuracy levels.

Gohr *et al.* demonstrated that the accuracy of a (classical) differential distinguisher is equivalent (up to constants) to the mean absolute distance between the ciphertext-difference distribution and the uniform distribution. They further established that for ciphers where a key-independent transformation can represent (intermediary) ciphertexts in the form $c' \oplus k'$ (with k' being independent key bits), the encryption distribution and ciphertext-difference distribution contain identical information [23]. This theoretical framework applies to major cipher classes: Substitution-Permutation Networks (SPNs, like Present, which alternate substitution and permutation operations) and Feistel ciphers (like Simon, which split data and apply functions to one half) both satisfy these conditions under the assumption of independent round keys (separate keys used in each encryption round)[2]. Consequently, neural distinguishers for these ciphers are fundamentally limited to learning differential features and can at best approximate the Difference Distribution Table (DDT). This limitation was empirically confirmed: Simon's neural distinguisher achieved virtually identical accuracy to a DDT-based distinguisher, and the Real-Difference Experiment showed Simon's accuracy dropped to exactly 0.5 (random guessing level), confirming exclusive reliance on differential features. Speck presents a contrasting case: its use of modular addition rather than XOR operations means it does not satisfy the theoretical constraints. Empirically, Speck's neural distinguisher consistently outperformed purely differential approaches and maintained significant accuracy (above 0.5) in the Real-Difference Experiment, demonstrating its ability to exploit non-differential features. Despite these theoretical limitations, neural differential distinguishers remain valuable tools for cryptanalysis. Computing the DDT is computationally expensive or infeasible for large block sizes, while neural distinguishers provide fast, accurate approximations obtainable within hours on consumer hardware.

Bao *et al.* developed explicit rules to be used alongside a differential distinguisher to enhance its effectiveness and more closely match the performance of advanced neural distinguishers [169]. The rules are based on strong correlations between bit values in the right pairs of XOR-differential propagation through addition modulo 2^n (how XOR differences behave when processed through modular addition operations). The authors also showed that those rules can be closely linked to the previous studies of the multi-bit constraints and the fixed-key differential probability. Finally, the authors concluded that leveraging the value-dependent differential probability makes it possible to add additional knowledge to purely differential distinguishers. In contrast, they demonstrate that neural differential distinguishers inherently utilize these rules. Building on this observation, Lv *et al.* [194] trained a neural distinguisher on differential-linear cryptanalysis.

The collective research trajectory from Gohr's initial experiments through subsequent investigations by Benamira et al., Bellini et al., Gohr et al., and Bao et al. reveals a progressive refinement in understanding how neural networks operate in cryptanalysis. This research demonstrates a clear evolution from observing that neural distinguishers for some class of ciphers exploit non-differential (key-independent) features to identifying specific differential-linear patterns they recognize, and finally to formalizing these insights into explicit rules based on bit-value correlations and value-dependent differential probability.

b) Understanding the network architecture: In [167], Bacuieti *et al.* investigated the structure of the neural network itself. In particular, the authors used the *lottery ticket hypothesis* (the idea that sparse subnetworks within larger networks can achieve comparable performance) to prune Gohr's neural network to a minimal working version, on which they used feature visualization techniques to obtain a visual representation of the neural network's behavior. They additionally show that, for the case of SPECK32, there is no significant accuracy difference between the depth 1 neural network and the depth 10 version for Speck reduced to 7 and 8 rounds.

c) Understanding the influence of the input data on the distinguisher performance: Ablation studies are routinely performed for neural networks to understand their sensitivity and fidelity under small perturbations on either the network itself or its input data. Ablation studies can give insights into the explainability of neural network models, as detailed, for example, in *"BASED-XAI: Breaking Ablation Studies Down for Explainable Artificial Intelligence"* [229], or *"Logic Rule Guided Attribution with Dynamic Ablation"* [230]. In [217], Yue *et al.* performed a data ablation study to observe trade-offs between improved accuracy and overfitting when using multiple ciphertext pairs per sample for neural differential distinguishers. Empirical evidence consistently demonstrates that training on limited sample sizes significantly increases the risk of overfitting, mostly independent of the number of pairs per sample.

Seok *et al.* [203] investigated the use of Principal Component Analysis (PCA) and K-means clustering to define metrics

[2]This assumption is empirically validated: The authors' experiments across six ciphers demonstrated that key schedule algorithms (methods for generating round keys from the main key) have negligible impact on distinguisher performance, as replacing cipher-specific round keys with independent, identically distributed keys produced no significant accuracy changes.

for evaluating the quality of datasets in differential-neural cryptanalysis. Their findings reveal that the datasets associated with input differences leading to successful distinguishers, which can effectively separate cipher outputs from random data, tend to have more axes that effectively represent the data compared to other datasets. Similarly, these datasets form multiple high-density clusters compared to only a single cluster in the shape of a sphere. They introduce an input difference search method based on PCA and K-means clustering that surpasses the efficiency and effectiveness of the greedy approach proposed in [17].

d) Understanding the importance of bits: Recent advances in neural distinguishers [176], [178], [182], [193], [190], [180], [173], [19], [225], [175] have demonstrated remarkable efficiency by operating on partial ciphertext information rather than complete outputs. These approaches have simultaneously advanced cryptographic interpretability methods through systematic identification of the most influential ciphertext bits. Chen *et al.* [173] introduced "Informative Bits" and Bit Sensitivity Testing, formally defining informative bits as ciphertext bits that effectively distinguish between a cipher and a pseudo-random permutation. They successfully maintained high distinguisher performance for SPECK32/64 while omitting 16 of 32 ciphertext bits through their novel testing methodology.

Hambitzer *et al.*'s [180] deep learning ensemble (NNBits) provided bit-profiling capabilities specifically designed for evaluating cryptographic (pseudo) random bit sequences. Their work notably contributed to explaining the accuracy obtained by Gohr's depth-1 neural distinguisher in round 6 for SPECK32/64 by providing a detailed bit-level analysis. Liu *et al.* [190] performed a comprehensive interpretability analysis exploring the relationship between neural distinguishers, truncated differentials, and advantage bits (the most informative bits for cryptanalysis). Their advantage bit search algorithm successfully truncated ciphertexts to just 8 bits while leveraging XOR differences to reduce training sample size requirements significantly.

Similarly, Ebrahimi *et al.* [176] presented a Partial Differential (PD) ML-distinguisher for SPECK32/64, achieving nearly identical accuracy with only 8 bits compared to full 32-bit distinguishers for six rounds of the cipher. Goi *et al.* [178] employed explainable AI techniques (LIME and SHAP, which are model explanation methods) to examine Gohr's neural distinguisher, revealing significant methodological differences: LIME effectively captures individual bit significance, while SHAP uniquely identifies important bit pairings in the ciphertext.

Seok [19] developed a specialized neural distinguisher for HIGHT that focuses exclusively on ciphertext bits produced by one of the two independent operations in the round function, demonstrating the viability of operation-specific analysis. Zhang *et al.* [225] extended neural cryptanalysis to AES-128, training distinguishers for the 2-round reduced cipher and additionally examining specific intermediate states between rounds 2 and 3. Their approach replaced full 16-byte state processing with specialized networks operating on just 2-byte segments while maintaining nearly identical accuracy. Huang *et al.* [182] train partial neural distinguishers through extended encryption and strategic decryption with zero-set subkey bits, and Li *et al.* [193] develop a sophisticated ensemble approach combining multiple student distinguishers, each strategically trained on input differences producing mostly distinct informative ciphertext bits.

Deng *et al.* introduced the attention mechanism [8] into the differential cryptanalysis on SPECK [175]. The authors used a visualization algorithm to demonstrate the effectiveness of the attention mechanism and further analyze the features extracted from the ciphertext by deep learning. With this visualization technique, the authors evaluate which bits the attention mechanism focuses on most, providing interpretability results.

This extensive body of research demonstrates that neural distinguishers can achieve remarkable efficiency and interpretability by identifying and focusing on the most informative ciphertext bits. By systematically isolating critical bit positions through techniques like bit sensitivity testing, advantage bit search, and attention mechanisms, researchers have drastically reduced computational requirements for key recovery attacks (attacks that aim to find the secret encryption key). These approaches not only improve attack efficiency but also provide valuable insights into cipher vulnerabilities at the bit level, establishing a foundation for more targeted (neural) cryptanalysis.

VII. Overview: Neural Aided Key Recovery Attacks

Gohr's work [17] marked a breakthrough in ML-based cryptanalysis, achieving high-accuracy neural distinguishers for 7-round SPECK32/64 and developing key recovery attacks (cryptanalytic attacks that aim to find the secret encryption key) for 11 and 12 rounds that rivaled or surpassed state-of-the-art manual techniques.

Since then, research has progressed in multiple directions, including applying the proposed key recovery algorithm to various cryptographic primitives [206], [212], [207], [217], [219], [224], proposing enhancements to the original algorithm [187], [173], [166], [222], [201], [181], [174], [194], exploring key recovery in alternative adversarial settings [186], [189], [169], and reducing the complexity of the attack by truncating the ciphertexts observed by the distinguishers [173], [182], [193], [19], [225]. We specifically highlight (†) papers that implement a (full) Bayesian attack (probabilistic key search methods) compared to those employing a simplified basic attack.

1) Key Recovery on Different Cryptographic Primitives: While the first neural-aided key recovery was performed on SPECK32/64, subsequent works applied the same or a simplified version of the attack to SPECK [224], SIMON [206], [212], [224], LBC-IoT [207], SLIM [207], SPECK [217], and PRESENT [219].

Zhang *et al.* [224]† achieved significant breakthroughs in differential-neural cryptanalysis by performing key recovery attacks on 13- and 14-round SPECK32/64, with the 14-round attack exhaustively searching through the final round's subkey. They also executed the first 17-round key recovery attack on SIMON32/64. Building upon Gohr's foundational work [17], the authors implemented knowledge distillation (a technique to create smaller, efficient neural networks from larger ones) to create dramatically smaller student networks featuring fixed-size convolutions and GlobalAveragePooling layers. These streamlined architectures substantially reduced computational demands during key recovery while maintaining attack effectiveness.

Tian and Hu [206] developed 7-9 round neural distinguishers for SIMON32/64 and achieved 15-round key recovery using a prepended differential, which is a probabilistic transition from an input difference to an output difference, combined with probabilistic neutral bits that represent specific bit positions that can be varied without affecting the differential pattern, followed by an exhaustive subkey search.

Teng *et al.* [207] demonstrated practical 8-round key recovery attacks on LBC-IoT using their 6-round neural distinguisher.

Wu *et al.* [212]† introduced a mixed-neural differential network for 12-round SIMON32/64 key recovery, achieving higher accuracy with increased complexity.

Yue and Wu [217] improved upon Gohr's work with a novel data format exploiting SPECK32/64's round function structure, enabling 8-round key recovery.

Zhu *et al.* [219] successfully executed an 8-round key recovery attack on PRESENT by extracting non-linear S-box features (characteristics of the cipher's substitution boxes, which are key nonlinear components) using randomly generated subkeys, demonstrating that neural networks trained on ciphertext differences substantially outperform those trained on raw ciphertext pairs for distinguishing and key recovery tasks.

2) Advancements of the Key Recovery: Several studies aimed to advance neural-aided key recovery by focusing on parameter selection [187], [173], [166], exploring variants of neutral bits in the prepended classical differential [166], [222], [201], and reducing data complexity (the amount of encrypted data needed for the attack) through precomputation [181], [194] and reducing encryption queries [174].

Lyu *et al.* exhaustively explored neural distinguishers for Bayesian key search and applied them to SIMECK32/64 [187]†. They obtained 8/9/10-round neural differential distinguishers and recovered penultimate and last round subkeys for 13/14/15-round SIMECK32/64 with low data and time complexity. Their findings revealed that key response profile regularity, which measures how consistently a neural distinguisher's output changes when decrypting with systematically varied incorrect keys, plays a crucial role and varies greatly among distinguishers, as does the number of neutral bits available for the distinguisher's prepended differential. Interestingly, the most accurate neural distinguisher did not necessarily achieve the best key recovery performance.

Chen *et al.* proposed a Neural-Aided Statistical Attack (NASA) with experiments on reduced-round SPECK32/64, DES, and Speck96/96 [173]†. Their theoretical estimates suggest breaking 10-round DES, surpassing Gohr's 8-round attack. When combined with a novel early stopping technique, neutral bits, and a Bayesian algorithm in the lines of [17], their method reduces both computational and data complexity compared to the original key recovery in [17].

Bao *et al.* introduced generalized neutral bits techniques and conditional neural differential cryptanalysis [166]†. They improved the success rate of deep learning-assisted key recovery attacks by considering neural distinguisher accuracies, round numbers, and classical differential paths spliced in front of neural distinguishers. They also explored data complexity aspects and achieved successful key recovery attacks on 13-round SPECK32/64 and 16-round SIMON32/64.

In [222]†, the authors improved SIMECK-32 attacks, enhancing the 15-round attack and launching the first practical 16- and 17-round key recovery attacks for SIMECK32/64. They extended their 12-round neural distinguisher with a 3-round differential and associated 14 deterministic NBs (neutral bits with deterministic behavior) and 2 SNBSs (sets of neutral bits that can be simultaneously complemented) identified through exhaustive search.

In [201]†, the authors implemented full key recovery on Simon32/64 using a distinguisher trained for polytopic differences (differences with multiple possible patterns). Unlike Gohr's attack [17], their approach doesn't rely on neutral bits but instead filters $(r+1)$-round ciphertext pairs conforming to the initial differential using an $(r+1)$-round neural distinguisher, selecting pairs producing the highest scores.

Hou *et al.* in [181]† leveraged key response profile periodicity (repeating patterns in how keys behave) to achieve key recovery using only a partial profile. This approach is particularly necessary for block ciphers with round key sizes significantly larger than 16 bits, such as SIMON$64/128$, ensuring feasible key response profile generation.

In [174], the authors proposed a data reuse strategy for distinguishers processing input sets of $n > 2$ ciphertext pairs. Their approach generates a large ciphertext set and forms subsets where each ciphertext pair appears in a limited number of subsets while maintaining sufficient distinction between subsets. Using this strategy, they applied neural distinguishers to perform 10-rounds and 11-rounds key recovery on Speck32/64 using NASA [173] and Bayesian Key Recovery, respectively.

Lv *et al.* [194]† used super-neutral bits (enhanced neutral bits with stronger properties) to decrease attack data complexity and a lookup table strategy to eliminate real-time neural distinguisher invocations, performing a practical 13-round key recovery on Speck using their novel differential-linear neural distinguishers.

3) Bit-Level Ciphertext Analysis: Recent work on neural distinguishers [173], [182], [193], [19], [225] offers promising approaches for reducing both computational and data requirements in key recovery attacks. These distinguishers can operate on partial ciphertext bits, suggesting that complete decryption may not be necessary for successful key recovery. For detailed explanations of approaches involving distinguishers on partial ciphertexts, see section VI on interpretability. Here, we focus specifically on key recovery applications.

Chen *et al.* [173]† trained student distinguishers (smaller, specialized neural networks) using only subsets of ciphertext bits for DES and SPECK while maintaining high accuracy. For SPECK32/64, they omitted 6 of the 32 ciphertext bits, identified through their novel Bit Sensitivity Test. These optimized distinguishers enabled subkey recovery in smaller subspaces (reduced search spaces for key bits), reducing attack complexity. The authors demonstrated a practical attack on SPECK32/64 and provided theoretical estimates for attacking 10-round DES and 14-round Speck96/96.

Li *et al.* [193] extended this work by developing an ensemble of student distinguishers, each trained on distinct input differences and ciphertext bit combinations. Their key insight revealed that varying input differences cause different ciphertext bits to become critical for distinction, affecting relevant key bits. Through their novel key sensitivity test, they partitioned the subkey space into independently solvable components, enabling practical key recovery against previously resistant large-state block ciphers: 18-round SIMON128, 14-round SIMON96, 14-round SIMON64, 12-round SPECK128, 10-round SPECK96, and 9-round SPECK64.

Huang *et al.* [182] introduce a novel neural differential cryptanalysis framework that substantially mitigates computational complexity in large-state block cipher key recovery. By implementing a parallelizable multi-stage approach with strategically trained neural distinguishers, the researchers demonstrate improvements in attacking SPECK. The proposed methodology leverages partial neural distinguishers (PNDs, networks trained on subsets of the cipher state) executed in parallel, followed by a full neural distinguisher (FND) for key selection. The partial distinguishers are trained to recover independent key bits through an innovative whitening key decryption technique (a method to isolate specific key bits during analysis). Experimental validation on 10-round SPECK64 and 10-round SPECK96 reveals computational efficiency gains. Their SPECK64 attack employed a customized ResNet architecture using multiple ciphertext pairs generated via neutral bits and an advanced staged training protocol.

Seok *et al.* [19] attempted partial key recovery in the final transformation of 15-round HIGHT, claiming to recover portions of the last round key. However, their analysis relied on an assumed differential characteristic with probability 2^{-31} without addressing practical implementation details, particularly regarding the use of this prepended differential characteristic without neutral bits. While they theorized about a divide-and-conquer strategy, the proposal lacked concrete implementation details.

Zhang *et al.* [225] developed neural distinguishers targeting a 2-round reduced version of AES-128, specifically analyzing pairs of bytes from the ciphertext. These byte-wise distinguishers were leveraged to mount key recovery attacks on 3-round AES-128 using a divide-and-conquer strategy where different key segments were recovered independently.

4) Key Recovery for Related-Key and Conditional Adversaries: Conditional and related-key differential cryptanalysis enhances adversarial capabilities by slowing the diffusion of differences, enabling attacks on additional rounds of ciphers. Related-key attacks assume the attacker can observe encryption under multiple related keys, while conditional attacks constrain specific plaintext or key bits. Following trends in classical cryptanalysis, this approach has extended key recovery attacks in the related-key setting for SPECK [169] and in conditional and related-key settings for KATAN [186], [189].

Lin *et al.* demonstrated practical key recovery attacks on KATAN ciphers [186]† by combining conditional and related-key differential cryptanalysis. They successfully attacked 125-round KATAN32, 106-round KATAN48, and 95-round KATAN64, while proposing parallelization of the Wrong Key Response Profile (a measure of how incorrect key guesses behave) calculation to enhance efficiency.

In subsequent work, Lin *et al.* developed attacks targeting 97-round KATAN32, 82-round KATAN48, and 70-round KATAN64 [189]†. Their method integrates neural distinguishers with conditional prepended differentials that constrain specific plaintext and key bits. By identifying optimal conditions and neutral bit sets, they improved the attack effectiveness.

Bao *et al.* [169] successfully executed a 14-round key recovery attack on SPECK32/64 in the related-key setting.

5) Limitations of Hybrid Distinguisher Models for Key Recovery: Recent works [216], [214], [213] have explored hybrid approaches combining classical differential transitions with neural distinguishers. Rather than optimizing standalone s-round neural distinguishers, these models are constrained to input differences matching the output difference of a classical r-round differential transition. This hybridization aims to achieve near-perfect distinction for $(r+s)$-encrypted ciphertexts with minimal data complexity.

A significant limitation of these hybrid approaches stems from their dependence on extended classical differential paths. This dependence precludes the use of neutral bits, making it impossible to construct the plaintext structures (see section IV) necessary for key recovery methods like those in [17]. While some studies, including [216], have claimed breakthrough results in key recovery, these claims lack experimental validation.

In a different approach, Yadav *et al.* [218] constructed high-accuracy neural distinguishers from low-accuracy ones without prepending classical differentials, though at the cost of increased data complexity.

VIII. Comparative Review: Neural Differential Distinguishers

In the following, we provide a comparative review of all trained neural differential distinguishers as of February 2025. First, all investigated neural network architectures are reviewed (subsection VIII-A), then we detail the classification scheme (subsection VIII-B) and conclude with a comparative review of the best neural differential distinguishers for each symmetric primitive (subsection VIII-C), followed by a discussion of the review (subsection VIII-D).

A. Architectures

In this section, we review the neural network architectures[3] employed in Neural Differential Cryptanalysis.

1) Core Architectures:

a) $\mathcal{ND}_{\text{Gohr}}$: $\mathcal{ND}_{\text{Gohr}}$ is the foundational neural network architecture introduced by Gohr [17]. The architecture consists of four key components: (1) input reshaping that mirrors the cipher's word-oriented structure, (2) a single bit-sliced convolution enabling effective learning of XOR relationships between components [165], (3) a residual convolutional tower of varying depths (commonly depth-1, 5, or 10), and (4) a fully connected prediction head. The bit-sliced convolution is particularly crucial, as other architectures struggle without direct XOR inputs [22], [179], [216]. Initially requiring elaborate staged training for 8-round SPECK (Table III) distinguishers, subsequent work demonstrated that freezing all layers except the prediction head enables rapid training [169]. $\mathcal{ND}_{\text{Gohr}}$ has been applied to the majority (14/24) of primitives and serves as the baseline for most cryptanalytic applications.

Several variants extend the original design: $\mathcal{ND}^{\text{pruned}}_{\text{Gohr}}$ reduces model complexity for SPECK (Table III) [167]; $\mathcal{ND}^{\text{attntn.}}_{\text{Gohr}}$ adds attention mechanisms to SPECK (Table III) [175]; $\mathcal{ND}^{\text{ensmbl.}}_{\text{Gohr}}$ combines multiple networks for enhanced explainability on SPECK (Table III) [180]; and $\mathcal{ND}^{\text{sep.conv.}}_{\text{Gohr}}$ employs separable convolutions to reduce training costs for SPECK (Table III) [190].

b) DBitNet: DBitNet was designed as a "cipher-agnostic" alternative to $\mathcal{ND}_{\text{Gohr}}$ [42]. Built on dilated convolutional layers that learn dependencies between distant rather than neighboring neurons, DBitNet eliminates the need for input reshaping and bit-slicing convolution. This design enables automatic distinguisher generation across multiple primitives while achieving comparable accuracy to Gohr's method through simple staged training. The architecture successfully generated distinguishers for eight primitives: SPECK (Table III), SIMON (Table I), HIGHT (Table XIII), PRESENT (Table XVIII), GIMLI (Table XI), KATAN (Table XIV), TEA (Table XXIII), and LEA (Table XVII).

c) Inception: The INC architecture incorporates GoogLeNet's Inception modules, which process inputs through multiple parallel convolutional layers with various kernel sizes. This parallel processing enables feature extraction capabilities that single kernel sizes cannot achieve, though at increased computational cost [23]. First proposed in [224], INC has been applied to SIMECK (Table II), PRESENT (Table XVIII), CHASKEY (Table VII), DES (Table VIII), SIMON (Table I), and SPECK (Table III). A variant $(\text{INC}^{\text{freeze}})$ employs staged training with partially frozen networks, treating convolutional layers as reusable feature extractors while updating only the classification layers for new rounds [169].

d) MLP: MLP (Multi-Layer Perceptron) employs densely connected layers where each neuron connects to all neurons in subsequent layers. Due to their computational efficiency, MLPs have been investigated on 11 of 24 primitives.

Other classical machine learning methods (AdaBoost, Random Forest, Decision Trees) generally achieve accuracy rates at least 20% lower than CNNs [221]. Classical ML methods like SVM are occasionally competitive and included when relevant [168].

e) SE-Networks: SENet (Squeeze-and-Excitation Network) introduces attention mechanisms that adaptively recalibrate channel-wise feature responses [166]. The core squeeze-and-excitation blocks perform: (1) squeeze operations using global average pooling to create channel descriptors, (2) excitation operations with fully connected layers learning channel dependencies, and (3) scale operations multiplying features by attention weights. This enables focus on discriminative cryptographic features with minimal overhead, proving particularly effective for longer encryption rounds on SIMECK (Table II) and SIMON (Table I).

SE-ResNet combines ResNet's residual connections with SENet's attention mechanism, leveraging both gradient flow improvement and adaptive feature recalibration [192]. Applied to SIMON (Table I) and SIMECK (Table II), this hybrid approach demonstrates the effectiveness of combining complementary architectural innovations.

2) Additional Architectures:

a) LSTM and Transformer: LSTMs (Long Short-Term Memory) process sequential data using memory cells to capture long-term dependencies in cryptographic patterns. Applied to GIMLI (Table XI), TinyJAMBU (Table XXIV), and GIFT (Table X) [22], [21], LSTMs treat ciphertext pairs as sequential data. Transformer architectures, based on attention mechanisms, process ciphertext pairs through encoder networks with positional encoding. Both architectures have been applied to LEA (Table XVII), PRESENT (Table XVIII), and HIGHT (Table XIII), offering alternative approaches to traditional convolutional methods [172].

b) UNet: UNet employs an encoding-decoding architecture typically used for image segmentation. The encoder compresses input features while the decoder reconstructs relevant patterns, connected by skip connections that preserve spatial information. Applied to GIFT (Table X) and PRESENT (Table XVIII) [219], UNet offers a different perspective on feature extraction by explicitly modeling the encoding-decoding process inherent in cryptographic analysis.

c) Quantum Neural Networks: Quantum Neural Networks leverage quantum computing principles, including superposition (qubits existing in multiple states simultaneously) and entanglement (correlated qubit interactions). Unlike classical networks processing binary information, quantum networks can theoretically explore multiple solution paths simultaneously. The first simulated quantum neural network distinguisher was demonstrated on SPECK (Table III) [184], though practical quantum implementations remain limited by current hardware constraints.

[3] We introduce key vocabulary for Neural Differential Cryptanalysis architectures: MLPs use *densely* connected layers with full connectivity between neurons, resulting in many parameters. *Convolutional layers* (CNNs) apply filters to detect spatial patterns, requiring more computation but better capturing hierarchical features. *Inception modules* combine parallel convolutions with various kernel sizes for enhanced feature extraction. *Residual connections* (RESNets) create bypass paths to improve information flow during training. *LSTM* (a type of RNN) processes sequential data using memory cells to capture long-term dependencies. *Attention mechanisms* dynamically focus on relevant input portions, forming the basis of transformer networks.

d) DenseNet: DenseNet implements dense connectivity where each layer receives inputs from all preceding layers through concatenation: $\mathbf{x}_\ell = H_\ell([\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_{\ell-1}])$. Organized into dense blocks connected by transition layers (1×1 convolutions and average pooling), the architecture employs a growth rate parameter k controlling feature map production. This design enables extensive feature reuse and improved gradient flow, which is beneficial for capturing multi-level cryptographic patterns. Applied to SPECK-32 (Table III), DenseNet showed marginal improvements over ResNet for shorter rounds but struggled with complex 8-round encryption [202]. While investigated in [166], it was surpassed by SENet for longer rounds and therefore does not appear in the following compilation of best neural distinguishers.

B. Classification Scheme n-m-T-E for Neural Distinguishers

The proliferation of diverse training configurations for neural distinguishers often complicates the comparison of results across studies. Bellini *et al.* [42] addressed this challenge by proposing a systematic classification framework based on four key parameters: n, m, T, and E. We adopt this classification scheme throughout our review due to its demonstrated robustness in organizing the extensive cryptographic literature, extending it with additional taxonomic symbols to enable more nuanced categorization. We note that [42] is a technical contribution paper focused on developing automated neural cryptanalysis tools, not a survey paper, and makes no claims regarding a systematic or comprehensive collection of research literature.

To illustrate this framework, consider Gohr's foundational approach, which exemplifies the 2-1-CT-R configuration: the distinguisher processes pairs of ciphertexts ($n = 2$), considers only a single input difference ($m = 1$), receives raw ciphertext pairs without preprocessing ($T = \mathrm{CT}$), and performs binary classification to detect the usage of the input difference ($E = \mathrm{R}$).

Parameter Overview of n-m-T-E Classification

- n: Number of ciphertexts per training sample
- m: Number of input differences per training sample
- T: Feature engineering on training sample
- E: Type of distinguishing experiment performed

1) Number of Ciphertexts per Sample: n: This section focuses on multiple samples ($n > 2$) for a single input difference ($m = 1$). The next section examines multiple input differences ($m > 1$).

The foundational work by Gohr [17] already demonstrated that, using multiple independent ciphertext pairs (generated from the same input difference δ), predictions from a single-pair distinguishers could be effectively combined during key recovery, amplifying signal strength and improving classification accuracy. This score aggregation approach was later formalized by Gohr *et al.* [23], who noted its rediscovery across multiple studies and proposed systematic combining formulas.

From a statistical perspective, using multiple independent samples to classify between two distributions is fundamental to hypothesis testing theory. When observing samples $X_1, X_2, \ldots, X_n$ from either distribution f_0 (null hypothesis) or f_1 (alternative hypothesis), the likelihood ratio test $\Lambda = \prod_i f_1(X_i)/f_0(X_i)$ provides optimal classification. Performance improves exponentially with sample size n, as both Type I and Type II error probabilities decay exponentially when distributions differ, a classical result established by Neyman and Pearson (1933).

Instead of combining prediction scores using systematic formulas, Shen *et al.* [204] replaced traditional score aggregation with multilayer perceptrons (MLPs) that directly classify from multiple pair scores. Further, researchers have moved beyond post-hoc score combination to architectures that process multiple ciphertext pairs jointly during inference. This approach was pioneered by Benamira *et al.* [165], whose distinguisher directly accepted multiple pairs as input, and has since been adopted across numerous subsequent works.

The benefits of combining multiple pairs during inference remain contentious. While joint processing could, in theory, allow neural distinguishers to exploit key schedule biases or plaintext correlation patterns, empirical evidence suggests limited advantages.

Key schedule influence appears minimal, as Gohr *et al.* [23] showed that replacing round keys with uniform random values negligibly affects distinguishing accuracy. Similarly, Chen *et al.* [174] explored two critical scenarios: samples where all pairs of a multi-pair distinguisher share the same encryption key versus samples with independent keys per pair. Key reuse patterns showed minimal impact on distinguisher performance, suggesting neural networks learn features independent of key correlations.

In general, multi-pair architectures historically show minimal practical gains over classical score aggregation [23], [174], while increasing computational costs by processing x-fold more bits. Further, Zhang *et al.* [223] identified critical training trade-offs when using more samples. Comparing fixed total pairs (10^7) versus fixed multi-pair samples revealed that maintaining pair count led to overfitting, validation fluctuations, and slow convergence due to reduced sample diversity. This finding, confirmed by Zhang *et al.* [224], emphasizes that the number of samples has to be significantly increased for the training of multi-pair distinguishers.

2) Number of Input Differences: m: Traditional differential cryptanalysis focuses on single input differences, but advanced approaches can leverage multiple differences simultaneously, as formalized in multiple differential cryptanalysis [231]. It has been shown that multiple differential cryptanalysis, which simultaneously exploits differentials with different input differences, reduces data complexity by approximately a factor of $|\Delta_0|$ compared to single-difference attacks, where $|\Delta_0|$ represents the number of distinct input differences used. Blondeau and Gérard's theoretical framework demonstrates that combining multiple differential paths extracts more statistical information per sample, enabling practical improvements such as extending differential attacks on PRESENT from 16 to 18 rounds while maintaining reasonable data requirements.

Mirroring this trend in Neural Differential Cryptanalysis, recent advances in neural differential cryptanalysis have systematically explored multiple input difference techniques across various approaches and primitives. Baksi *et al.* [22] demonstrated the effectiveness of multi-difference settings with $m = 2$ across permutations including KNOT, ASCON, CHASKEY, and GIMLI, while two works ([201], [21]) adapted polytopic cryptanalysis to develop neural differential networks that maintain a fixed plaintext anchor while using multiple differences $(P, P \oplus \delta_0, P \oplus \delta_1, \ldots)$ to generate k differential pairs from $k + 1$ plaintexts. Similarly, Wu *et al.* [212] explored the combination of mixture differential techniques and machine learning. In [211], the authors investigated two variations of a multiple-input differences scenario, where the samples are the concatenations of pairs with differences δ_i. In ND_{rm}, a sample is the concatenation of a pair of ciphertexts for each difference (resulting in $n = 2m$); in ND_{am}, the first ciphertext is the encryption of a random plaintext P_0, each subsequent ciphertext C_i is the encryption of $P_{i-1} \oplus \Delta_{i-1}$ so that $n = m + 1$. Multiple works ([215], [22], [168], [199], [210]) trained neural distinguishers to differentiate between different output difference distributions (see subsubsection VIII-B4).

3) Feature Engineering Type: T:

a) Raw Ciphertext Processing (T = CT): The most straightforward feature engineering type passes the raw ciphertext pairs directly to the neural distinguisher. While this preserves maximum information content, it requires networks to learn differential patterns implicitly, potentially increasing training and architectural complexity [165].

b) XOR Difference Features ($T = \delta$): Multiple works, such as Baksi *et al.* [22], Hou *et al.* [179], Yadav *et al.* [216], have demonstrated that replacing ciphertext pairs with their XOR differences can speed up training while maintaining comparable performance. This transformation reduces input dimensionality and provides a more direct representation of differential characteristics.

Truncated Variants ($T = \mathrm{CT}_{\mathbf{tr}}/\delta_{\mathbf{tr}}$): Huang *et al.* [182] and Yhang *et al.* [225] reduced training requirements by truncating ciphertexts to a few bits before handing them to the distinguisher, significantly decreasing sample size requirements at the cost of information loss. Liu *et al.* [190], Ebrahimi *et al.* [176], and Seok [19] further reduced training requirements by truncating ciphertexts to a few bits before computing XOR differences, decreasing sample size requirements at the cost of information loss.

Advanced Variants ($T = A$): In [170], the authors append the individual ciphertext pairs and the ciphertext difference.

For **Partial Decryption Without Key Knowledge**, researchers demonstrated that XORing cipher halves and applying rotation operations reveals structural information that improves distinguisher accuracy for SPECK32 [165], [182], [224], with similar feature engineering applied to SIMON-like ciphers by Bao *et al.* [166]. However, Gohr *et al.* [23] showed that neural distinguishers can learn these feature transforms automatically when provided with raw ciphertext pairs, though this requires extensive hyperparameter optimization, demonstrating how the cipher structure influences the optimal architecture choice of the neural distinguisher.

For **Partial Decryption With Key Assumption**, Yue *et al.* [217] developed the format $(R_{r-1}, R'_{r-1}, d_l, C_0, C_1)$ where d_l estimates the left-half difference at round $r - 1$ through $((L_r \boxminus R_{r-1}) \oplus (L'_r \boxminus R'_{r-1}))$, equivalent to partial decryption with a null key. Lu *et al.* [192] extended this by incorporating two-round prior information using subkey 0 for partial decryption, an approach also used in [215], [222]. The **MRMSD/MRMSP Approaches** by Liu *et al.* [191] introduced MRMSD (Multiple Rounds Multiple Splicing Differences), combining output differences with estimated previous-round differences using random-key partial decryption, and MRMSP (Multiple Rounds Multiple Splicing Pairs), providing corresponding ciphertext pairs instead of differences. Finally, Zhu *et al.* [219] modified traditional approaches by performing partial encryption to intermediate round states, targeting input/output relationships of substitution box operations to extract nonlinear transformation features.

Feature Engeneering T Overview

- CT: Unprocessed ciphertext pair handed to $\mathcal{ND}$
- δ: Ciphertext XOR handed to $\mathcal{ND}$
- $\delta_{\mathbf{tr}}$: Reduced-bit XOR differences handed to $\mathcal{ND}$
- A: Advanced processing, such as partial decryption

4) Type of Distinguishing Experiment: E:

a) Standard Differential Detection (E = R): Gohr's [17] foundational experiment uses samples of the form $E_K(P_0) \| E_K(P_0 \oplus x)$ where the network determines whether x equals a target difference δ. This binary classification task directly mirrors classical differential cryptanalysis objectives.

b) Real Ciphertext Masking ($E = \mathrm{R_M}$): Gohr's [17] "real ciphertext" experiment creates samples $E_K(P_0) \oplus x \| E_K(P_0 \oplus \delta) \oplus x$ where the network identifies whether $x = 0$ (unmasked) or x is random (masked). Success in this experiment demonstrates that neural distinguishers learn information beyond simple XOR differences, capturing more subtle structural patterns.

c) Alternative Difference Operations ($E = \mathrm{R}^+$): **Modular Addition Differences**: Bellini *et al.* [20] adapted the framework for ciphers using modular addition (TEA, RAIDEN) by replacing XOR with modular addition differences, demonstrating the framework's flexibility across different algebraic structures.

Rotational-XOR Differences: Ebrahimi *et al.* [177] explored rotational-XOR differences rather than standard XOR operations, targeting ciphers with rotational components.

d) Multi-Difference Classification (E = D): Several works, including Baksi *et al.*'s [22], including Baksi *et al.*'s [22], Wang *et al.*'s [210], and Wang *et al.*'s [214], transform the binary classification problem into multi-class classification. Samples are formed as $(E_K(P), E_K(P \oplus \delta_i))$ for $i \in [0, m-1]$, with the network identifying which specific difference i was used. This approach enables simultaneous analysis of multiple differential characteristics.

e) Differential-Linear Hybrid Approaches (E = DL)): Lv *et al.* [194] developed a hybrid methodology that combines differential and linear (neural) cryptanalysis. For each ciphertext pair (c, c'), they apply N distinct output masks $(\Gamma_1, \Gamma'_1), \ldots, (\Gamma_N, \Gamma'_N)$ to generate N-bit input vectors where each bit computes $x_i = \Gamma_i \cdot C_1 \oplus \tilde{\Gamma}_i \cdot C_2$. This approach effectively implements differential-linear cryptanalysis within the neural framework, leveraging both differential properties and linear approximations.

Experiment Type E Overview

- R: $E_K(P) \| E_K(P \oplus x)$, Classify $x \stackrel{?}{=} \delta$
- $\mathrm{R_M}$: $E_K(P_0) \oplus x \| E_K(P_0 \oplus \delta) \oplus x$, Classify $x \stackrel{?}{=} 0$
- R^+: $E_K(P) \| E_K(P \circledast x)$, Classify $x \stackrel{?}{=} \delta$, where $\circledast \in \{\boxplus, \ggg\}$
- D: $(E_K(P) \| E_K(P \oplus \delta_i))$, Identify $i \in [0, m-1]$
- DL: $E_K(P) \| E_K(P \oplus x)$, Extract $\mathbf{b} = b_1 \| \cdots \| b_N$ where $b_i = \Gamma_i \cdot C_0 \oplus \bar{\Gamma}_i \cdot C_1$, then classify $x \stackrel{?}{=} \delta$

This classification framework provides a systematic foundation for comparing neural distinguisher approaches across the literature. The parameter space defined by n-m-T-E enables researchers to precisely characterize their methodologies and identify gaps for future exploration.

C. Comparative Review

Based on the whole body of research in Neural Differential Cryptanalysis (subsection V-A), this section provides a comparative review of all best published neural distinguishers, classified according to the previously introduced scheme, together with their neural network architecture (subsection VIII-A).

The neural differential distinguishers of each publication were selected as follows: *i)* We present the *best* result of each work, either the standard setting (2-1-CT-R or 2-1-δ-R) or an alternative setting (n-m-T-E). If, additionally, a result in the standard setting is given, we will also present it. *ii)* In most works, no error margins on the results are provided, preventing us from displaying them. Ideally, the accuracies shown should be test accuracies on sets of several fresh samples. However, in many works, only the validation accuracy is reported. *iii)* Note that from a machine learning and statistical perspective, the number of training and validation samples is very important. However, from a cryptographic perspective, the number of needed encryptions, i.e., ciphertexts, is more relevant. Accordingly, the numbers reported in the following under **Trn.** (training data) and **Val.** (validation data) are the number of *ciphertexts*.

To date, neural distinguishers have been applied to analyze 24 symmetric primitives. This comparative review provides a foundation for integrating new research into the existing body of work. While comprehensive tables compiling these analyses are provided in the appendices, we focus here on the most extensively studied primitives where distinguishers have been developed by 9 or more works (namely SIMECK, SIMON, and SPECK), as only these provide sufficient data for meaningful comparative analysis of optimal approaches. The complete list of primitives analyzed to date is as follows: AES (Table IV), ARADI (Table V), ASCON (Table VI), CHASKEY (Table VII), DES (Table VIII), FF (Table IX), GIFT (Table X), GIMLI (Table XI), GOST (Table XII), HIGHT (Table XIII), KATAN (Table XIV), KNOT (Table XV), LBCIoT (Table XVI), LEA (Table XVII), PRESENT (Table XVIII), PRIDE (Table XIX), SHA3 (Table XX), SIMECK (Table II), SIMON (Table I), SKINNY (Table XXI), SLIM (Table XXII), SPECK (Table III), TEA and XTEA (Table XXIII), and TinyJAMBU (Table XXIV).

In addition, Bose *et al.* claimed statistically significant distinguishers for 6-round SPARX and 8-round PICCOLO-80. As theirs is the only work targeting these ciphers and their reported improvements for other ciphers challenge fundamental cryptographic principles, we omitted dedicated tables that would lack contextual comparison. Instead, we included their distinguishers for LEA, PRESENT, and HIGHT with a critical discussion.

1) SIMON: SIMON is a family of AND-RX block ciphers, denoted SIMON-B/K, that encrypt blocks of size B with a key of size K. SIMON-32/64, SIMON-48/96, SIMON-64/128, and SIMON-128/256 have 32, 36, 44, and 72 rounds, respectively. Neural differential distinguishers have been developed for all versions of SIMON.

Table I provides an overview of the differential neural distinguishers developed for the SIMON family of block ciphers. The most extensively studied variant is SIMON-32, with various neural network architectures and settings explored across multiple works. In the standard setting (2-1-CT-R), the best distinguisher achieves round 11 through an automated pipeline and a generic, convolutional architecture [42]. In general, convolutional architectures consistently achieve the highest accuracy among the neural network distinguishers obtained for SIMON. By using multiple ciphertext pairs ($n = 16, 32, 64$) and employing advanced feature engineering techniques, as in [191], [192], [224], the distinguisher performance surpasses this result, extending the analysis to round 12 [192]. Similarly, switching to the related-key setting, distinguishers can be trained for 13 rounds, as in [192]. The classification task becomes significantly easier when distinguishing between two ciphertext distributions rather than distinguishing a single distribution from uniform[4]. This approach yields substantially higher accuracies in the related-key setting and enables attacks on one additional round using just a single ciphertext pair in the standard setting [215].

For the case of SIMON, some authors experimented with a vast amount of data: [179] used $k \cdot 10^7$ for $k = 32, 48, 64$ (maximum of 640M) samples for training, and [166] obtained an 11-round distinguisher for SIMON32 at the cost of staged trained in four steps, with respectively 10^7, 2^{28}, $2 \cdot 2^{30}$ (2426M pairs). In [42], the authors proposed a polishing step, retraining a neural distinguisher initially trained with 10^7 pairs with an additional 10^9 pairs: 10^7, 10^9 (1010M pairs). This polishing step was also used by Wang et al. [214]. Similarly, Zhang *et al.* [224] used a staged training approach: $4 \cdot 10^7$ samples,

[4]This approach is aimed to maximize the average absolute distance between two ciphertext distributions, a key metric for distinguisher performance, as established by Gohr [23]

TABLE I
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR SIMON.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
SIMON-32/64	$\mathcal{ND}_{\text{Gohr}}$	2-1-A-R	20M	2M	-	8	0.834	[165]
	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT-R	20M	2M	-	9	0.5907	[179]
	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT-R	20M	2M	-	9	0.6277	[201]
	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT-R	/	/	-	9	0.6320	[206]†
	$\mathcal{ND}_{\text{Gohr}}$	4-3-CT-R	40M	4M	-	9	0.6373	[201]
	$\mathcal{ND}_{\text{Gohr}}$	4-3-CT-R	40M	4M	-	8	0.923	[212]
	$\mathcal{ND}_{\text{Gohr}}$	64-1-δ-R	640M	6.4M	-	10	0.6109	[179]
	SENet	2-1-A-R	4852M	537M	-	11	0.517	[166]
	DBitNet	2-1-CT-R	2020M	2M	✓	11	0.518	[42]
	$\mathcal{ND}_{\text{Gohr}}$	64-1-A-R	640M	64M	-	11	0.6081	[191]
	DenseNet	2-2-CT-D	2020M	2M	✓	12	0.505	[214]
	SE-ResNet	16-1-A-R	320M	32M	-	12	0.5152	[192]
	INC	32-1-A-R	1280M	2M	-	12	0.5218	[224]
SIMON-32/64^{RK}	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT-R^{+}	20M	2M	-	11	0.5445	[177]
	SE-ResNet	16-1-A-R	320M	32M	-	13	0.5262	[192]
	SE-ResNet	16-2-A-D	320M	32M	✓	13	0.567	[215]
SIMON-48/96	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT-R	20M	2M	-	10	0.5789	[179]
	$\mathcal{ND}_{\text{Gohr}}$	96-1-δ-R	960M	9.6M	-	11	0.6143	[179]
	DenseNet	2-2-CT-D	20M	2M	✓	12	0.515	[214]
	$\mathcal{ND}_{\text{Gohr}}$	96-1-A-R	960M	96M	-	12	0.6159	[191]
SIMON-48/96^{RK}	SE-ResNet	16-2-A-D	320M	32M	✓	13	0.696	[215]
SIMON-64/128	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT-R	20M	2M	-	11	0.5972	[179]
	$\mathcal{ND}_{\text{Gohr}}$	128-1-δ-R	1280M	12.8M	-	12	0.6957	[179]
	DBitNet	2-1-CT-R	20M	2M	✓	13	0.518	[42]
	$\mathcal{ND}_{\text{Gohr}}$	128-1-A-R	1280M	128M	-	13	0.701	[191]
	DenseNet	2-2-CT-D	20M	2M	✓	14	0.506	[214]
	SE-ResNet	16-1-A-R	1394M	134M	-	14	0.5185	[192]
SIMON-64/128^{RK}	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT-R^{+}	20M	2M	-	13	0.5151	[177]
	SE-ResNet	16-1-A-R	320M	32M	-	14	0.5788	[192]
	SE-ResNet	16-2-A-D	320M	32M	✓	14	0.618	[215]
SIMON-128/256	DBitNet	2-1-CT-R	20M	2M	✓	20	0.507	[42]
SIMON-128/256^{RK}	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT-R^{+}	20M	2M	-	16	0.5062	[177]

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
/ Unknown quantity.
† A critical discussion of these results is provided in the text.
RK Related key setting.

each sample with 16 pairs (640M pairs). Lu *et al.* [192] use different training strategies depending on the scenario. In some cases, they employ a staged approach with $8 \cdot (2 \cdot 10^7 + 2 \cdot 2^{25})$ pairs, while in others they use $2 \cdot 10^7 \cdot 8$ pairs directly.

We end this section on SIMECK with a critical discussion of [206]. Tian *et al.* [206]† do not specify the size of their training or validation sets, making it impossible to assess the statistical reliability of their reported accuracy.

2) SIMECK: SIMECK is a variant of SIMON using a key schedule similar to that of SPECK. SIMECK-32/64, SIMECK 48/96, and SIMECK-64/128 have 32, 36, and 44 rounds, respectively. Neural differential distinguishers have been developed for all versions of SIMECK.

Table II provides an overview of the differential neural distinguishers developed for the SIMECK family of block ciphers. The most extensively studied variant is SIMECK-32, with various neural network architectures explored across multiple works. In the standard setting (2-1-CT-R), the best distinguisher achieves round 10 through the original convolutional architecture [187]. In general, convolutional architectures consistently achieve the highest accuracy among the neural network distinguishers obtained for SIMECK. By using multiple ciphertext pairs ($n = 16$) and employing advanced feature engineering techniques, as in [192], [222], [215], the distinguisher performance surpasses this result, extending the analysis to round 12 [192]. In the related-key setting, distinguishers can be trained for up to 15 rounds, a substantial improvement over the single additional round achieved for SIMON [192]. The task becomes easier when distinguishing between two ciphertext distributions rather than distinguishing a single distribution from uniform. This approach enables attacks on one additional round using just a single ciphertext pair in the standard setting [215].

For the case of SIMECK, some authors experimented with a vast amount of data: In [222], the authors used the staged training approach proposed by Gohr in [17]: $2 \cdot 10^7$, $2 \cdot 10^9$ samples with 8 pairs each (16160M pairs). In [42], the authors proposed a polishing step, retraining a neural distinguisher initially trained with 10^7 pairs with an additional 10^9 pairs: 10^7, 10^9 (1010M pairs). This polishing step was also used

TABLE II
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR SIMECK.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
SIMECK-32	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R	20M	2M	-	10	0.5407	[188]†
	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R	20M	2M	-	10	0.5438	[187]
	$\mathcal{ND}_{\mathrm{Gohr}}$	4-3-CT-R	67M	1M	-	11	0.5042	[211]
	DenseNet	2-2-CT-D	2020M	2M	✓	12	0.505	[214]
	SE-ResNet	16-1-A-R	1394M	134M	-	12	0.5146	[192]
	INC	16-1-A-R	32320M	32M	-	12	0.5161	[222]
SIMECK-32/64$^{\mathrm{RK}}$	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R$^+$	20M	2M	-	15	0.5134	[177]
	SE-ResNet	16-1-A-R	320M	32M	-	15	0.5467	[192]
	SE-ResNet	16-2-A-D	320M	32M	✓	15	0.568	[215]
SIMECK-32$^{\mathrm{Unkeyed}}$	MLP	2-2-δ-D	66K	66K	-	9	0.526	[168]‡
SIMECK-48/96	DenseNet	2-2-CT-D	20M	2M	✓	15	0.505	[214]
SIMECK-48/96$^{\mathrm{RK}}$	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R$^+$	20M	2M	-	17	0.5206	[177]
	SE-ResNet	16-2-A-D	320M	32M	✓	19	0.523	[215]
SIMECK-64/128	DenseNet	2-2-CT-D	20M	2M	✓	18	0.507	[214]
	SE-ResNet	16-1-A-R	1394M	134M	-	18	0.5218	[192]
	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R$^+$	20M	2M	-	20	0.5212	[177]
SIMECK-64/128$^{\mathrm{RK}}$	SE-ResNet	16-1-A-R	320M	32M	-	22	0.5180	[192]
	SE-ResNet	16-2-A-D	320M	32M	✓	22	0.526	[215]
SIMECK-64$^{\mathrm{Unkeyed}}$	MLP	2-2-δ-D	66K	66K	-	14	0.55	[168]‡

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
† A critical discussion of these results is provided in the text.
RK Related key setting.
‡ The use of a small validation set raises concerns about the statistical robustness, reproducibility, and generalizability of the results, as such datasets are prone to high variance and may not reliably reflect model performance.

by Wang et al. [214] Lu *et al.* [192] use different training strategies depending on the scenario. In some cases, they employ a staged approach with $8 \cdot (2 \cdot 10^7 + 2 \cdot 2^{25})$ pairs, while in others they use $2 \cdot 10^7 \cdot 8$ pairs directly.

We end this section on SIMECK with a critical discussion of [188]. In [188]†, the authors proposed training a neural distinguisher multiple times independently and selecting the model with the highest test accuracy. Notably, they reported successfully obtaining a single 10-round Simeck distinguisher in only one out of 20 independent training attempts.

3) SPECK: SPECK is a family of ARX block ciphers, denoted as SPECK-B/K, designed to encrypt blocks of size B with a key of size K. The variants SPECK-32/64, SPECK-48/96, SPECK-64/128, SPECK-96/96, and SPECK-128/256 consist of 22, 23, 27, 29, and 34 rounds, respectively. SPECK is the cipher with the most published neural distinguishers to date. Neural differential distinguishers have been developed for all SPECK versions, with a comprehensive overview presented in Table III.

In the standard setting (2-1-CT-R) for SPECK-32, Gohr's original analysis on 8 rounds [17] remains unmatched, in which the author applied a staged training: $2 \cdot 10^7$, $2 \cdot 10^9$ (2020M pairs). However, [42] achieved comparable accuracy to [17] using an automated, generic pipeline that is not specifically tailored to SPECK[5] and lacks the complex training scheme essential for high accuracy on 8 rounds. More precisely, the authors proposed a polishing step, retraining a neural distinguisher initially trained with 10^7 pairs with an additional 10^9 pairs (1010M pairs in total). Zhang et al. and Huang et al. [224], [182] used a staged training approach: $4 \cdot 10^7$ samples, each sample with 16 pairs (640M pairs). Wang and Wang [214] built upon the staged training proposed in [42]. As the size of the validation set has not been explicitly mentioned by the authors, we assume that they follow the size in [42].

Enhanced accuracy over Gohr's results on $\geq$ 8-round SPECK-32 can be achieved by employing advanced feature engineering and multiple ciphertext pairs (e.g., $n > 2$). Indeed, for 8-round SPECK-32, a higher accuracy is reported when using multiple ciphertext pairs for: [191] uses $n = 128$, and [194] uses $n = 512$. For 9-round SPECK-32, [224] presents the first distinguisher using 16 ciphertext pairs and advanced feature engineering.

The classification task becomes significantly easier when distinguishing between two ciphertext distributions rather than distinguishing a single distribution from uniform. This approach enables attacks on 8-round SPECK-32 with slightly higher accuracy in the standard setting [214].

[5]We note that [169] stated that "the simple training pipeline [of [42]] did not produce $\mathcal{ND}$s with the same accuracy as Gohr's on 8-round Speck32/64; it needs a further polishing step to achieve similar accuracy, demanding more time and data" which is not entirely correct. While in [42], a polishing step is indeed needed to achieve the same accuracy, the polishing step is a *highly simplified and automated* version of the 8-round training scheme used by Gohr (in conclusion, it does *not* demand more time or data).

TABLE III
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR SPECK.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
SPECK-32	Quantum	2-1-CT-R	60K	2K	-	5	0.53	[184]†
	$\mathcal{ND}^{\text{sep.conv.}}_{\text{Gohr}}$	2-1-δ_{tr}-R	10M	1M	-	6	0.673	[190]
	MLP	2-1-δ_{tr}-R	20M	2M	-	6	0.688	[176]
	MLP	2-1-δ-R	20M	2M	-	6	0.72	[176]
	$\mathcal{ND}^{\text{ensmbl.}}_{\text{Gohr}}$	2-1-CT-R	20M	2M	-	6	0.781	[180]
	$\mathcal{ND}_{\text{Gohr}}$	100-1-A-R	20M	2M	-	6	1	[165]
	DenseNet	2-1-CT-R	2M	2M	-	7	0.531	[202]†
	$\mathcal{ND}^{\text{pruned}}_{\text{Gohr}}$	2-1-CT-R	20M	2M	-	7	0.596	[167]†
	$\mathcal{ND}_{\text{Gohr}}$	2-1-δ-R	20M	2M	-	7	0.583	[165]
	$\mathcal{ND}_{\text{Gohr}}$	2-2-δ-D	20M	2M	-	7	0.599	[210]
	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT-R	2M	/	✓	7	0.614	[209]
	$\mathcal{ND}^{\text{attntn.}}_{\text{Gohr}}$	2-1-CT-R	20M	2M	-	7	0.6169	[175]
	$\mathcal{ND}^{\text{sep.conv.}}_{\text{Gohr}}$	8-1-CT-R	80M	8M	-	7	0.6939	[190]
	$\mathcal{ND}_{\text{Gohr}}$	16-1-CT-R	20M	2M	-	7	0.7009	[174]
	$\mathcal{ND}^{\text{attntn.}}_{\text{Gohr}}$	16-1-CT-R	160M	16M	-	7	0.728	[175]
	INC	64-1-A-R	64M	6.4M	-	7	0.9713	[217]
	$\text{INC}^{\text{freeze}}$	2-1-CT-R	20M	2M	-	8	0.5135	[169]
	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT-R	4040M	2M	-	8	0.514	[232]
	DBitNet	2-1-CT-R	2020M	2M	✓	8	0.514	[42]
	DenseNet	2-2-CT-D	2020M	2M	✓	8	0.519	[214]
	MLP	2-1-CT-DL	20M	2M	-	8	0.5208	[194]
	$\mathcal{ND}_{\text{Gohr}}$	128-1-A-R	1280M	128M	-	8	0.6502	[191]
	MLP	512-1-CT-DL	20M	2M	-	8	0.8866	[194]
	INC	32-1-A-R	1280M	2M	-	9	0.5045	[224]
$\text{SPECK-32}^{\text{RK}}$	$\mathcal{ND}_{\text{Gohr}}$	2-2-δ-D	20M	2M	-	7	0.559	[210]
	$\mathcal{ND}_{\text{Gohr}}$	2-2-CT-R	20M	2M	-	7	0.576	[210]
	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT-R	2M	200K	-	9	0.5932	[208]
	$\text{INC}^{\text{freeze}}$	2-1-CT-R	20M	2M	-	10	0.5562	[169]
$\text{SPECK-32}^{\text{Unkeyed}}$	MLP	2-2-δ-D	66K	66K	-	8	0.515	[168]‡
SPECK-48	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT-R	2M	/	✓	6	0.726	[209]
	DenseNet	2-2-CT-D	20M	2M	✓	8	0.506	[214]
	$\mathcal{ND}_{\text{Gohr}}$	128-1-A-R	1280M	128M	-	8	0.5462	[191]
SPECK-64	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT_{tr}-R	20M	2M	-	6	0.662	[182]
	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT-R	20M	2M	-	6	0.754	[182]
	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT-R	20M	2M	-	7	0.623	[182]
	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT-R	2M	/	✓	7	0.632	[209]
	INC	32-1-A-R	1280M	2M	-	7	0.641	[182]
	DBitNet	2-1-CT-R	20M	2M	✓	8	0.537	[42]
	DenseNet	2-2-CT-D	20M	2M	✓	8	0.559	[214]
	$\mathcal{ND}_{\text{Gohr}}$	128-1-δ-R	1280M	12.8M	-	8	0.632	[179]
	$\mathcal{ND}_{\text{Gohr}}$	128-1-A-R	1280M	128M	-	8	0.7181	[191]
SPECK-96	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT_{tr}-R	20M	2M	-	7	0.681	[182]
	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT-R	20M	2M	-	7	0.832	[182]
	$\mathcal{ND}_{\text{Gohr}}$	2-1-CT-R	20M	2M	-	7	0.850‡	[173]
SPECK-128	DBitNet	2-1-CT-R	20M	2M	✓	10	0.593	[42]

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).

/ Unknown quantity.

† A critical discussion of these results is provided in the text.

RK Related key setting.

‡ The use of a small validation set raises concerns about the statistical robustness, reproducibility, and generalizability of the results, as such datasets are prone to high variance and may not reliably reflect model performance.

‡ In [173], the accuracy of the teacher network for SPECK-96 was not given, but we were able to retrieve it by running the model from the authors' repository; we give the average of 10 runs, each with 10^6 samples.

† In [167], the authors evaluated several pruned neural distinguishers; we report the smallest one, Gohr's $\mathcal{ND}_{\text{Gohr}}$ with depth 1, 7 channels removed from C1, 21 from C2, 25 from C3, 46 neurons from D1, and 36 from D2.

For 8-round SPECK-32, Lv et al. [194] demonstrated that differential-linear cryptanalysis can produce neural distinguishers surpassing the state-of-the-art distinguishers based solely on differential cryptanalysis. Their methodology involved an exhaustive search over differential-linear approximations with low Hamming weights, filtering out the most influential approximations using importance metrics from the Light Gradient Boosting Machine (LGBM) classification algorithm. However, the authors incorrectly asserted that multi-pair neural differential distinguishers necessitate architectural modifications. This assertion directly contradicts the work of Gohr [23], which explicitly presented a concrete methodology for constructing multi-pair distinguishers from single-pair architectures without requiring structural changes. Notably, the input difference employed in their 8-round distinguisher exhibits significant divergence from conventional patterns established in current literature, raising interesting questions about optimal difference selection in neural cryptanalysis.

In general, convolutional architectures consistently achieve the highest accuracy among neural network distinguishers for SPECK, with the exception of results in [194]. However, since only MLPs have been trained for neural differential-linear cryptanalysis, it remains unclear whether convolutional neural networks would be superior in this setting as well.

Several studies demonstrate that neural distinguishers can achieve comparable performance using significantly truncated ciphertext data. Liu *et al.* [190] conducted interpretability analysis linking neural distinguishers to truncated differentials and advantage bits, developing an algorithm that successfully reduced ciphertexts to 8 bits while exploiting XOR differences to minimize training requirements. Ebrahimi *et al.* [176] presented a Partial Differential ML-distinguisher for SPECK32/64, achieving similar accuracy with 8-bit across six rounds. Huang *et al.* [182] similarly trained partial neural distinguishers on 11 ciphertext bits.

We end this section on SPECK with a critical discussion of [202] and [184]. Sajwan *et al.* [202]† reported an accuracy of 53.1% (round 7) on 2M training, respectively validation samples, and provides a comparison in which DenseNet outperforms $\mathcal{ND}_{\mathrm{Gohr}}$. At such a small number of training samples, both networks show heavy overfitting ([202, Table 2]), and the authors themselves called the result only "marginal." In [184]†, the author reported an accuracy of 53% (round 5) on only 1,000 validation samples. The experimental mean or standard deviation was not given. The statistically expected standard deviation for a binomial experiment on 1k samples is $1/(2\sqrt{n}) = 1.6\%$. Therefore, the reported result is only 1.9σ away from random and is likely not statistically significant.

D. Discussion

Based on our comprehensive assessment of research in neural differential cryptanalysis (subsection V-A), we identify several promising directions and critical challenges that merit further investigation. Our analysis focuses primarily on thoroughly vetted cryptographic primitives – those subjected to substantial cryptanalytic scrutiny, specifically SIMON (Table I), SIMECK (Table II), SPECK (Table III).

a) Network Architectures (N): Findings on optimal neural network architectures for cryptanalysis remain contradictory. While Baksi *et al.* [22] concluded CNNs were unsuitable for distinguishers and found MLPs superior on GIMLI-PERMUTATION, Bellini *et al.* [20] and Wang *et al.* [210] demonstrated effective CNN-based distinguishers for PRESENT and SPECK. Mishra *et al.* [195] reported MLPs outperforming CNNs on GIFT and PRIDE, whereas Sun *et al.* [21] found LSTMs superior to MLPs on TinyJAMBU and GIFT. Tcydenova *et al.* [208] evaluated various architectures but found no significant improvements over ResNet, though they noted overfitting issues. Lv *et al.* [194] comprehensively compared multiple techniques for differential-linear cryptanalysis, with MLPs consistently outperforming alternatives, including ELLR, Logistic Regression, and LightGBM.

Convolutional neural networks, pioneered as the original distinguisher architecture in [17], consistently demonstrate excellent performance in neural cryptanalysis, with convolutional architectures ranking among the most effective distinguishers across virtually all cryptographic primitives with a substantial body of neural cryptanalytic research. This is expected as DCNNs have demonstrated remarkable feature extraction capabilities across various disciplines, particularly in image recognition. Once the relevant features have been identified, classical or simpler neural models can often achieve performance comparable to their complex neural counterparts [165], [169]. However, meaningful comparison between concrete approaches remains challenging due to numerous influential factors beyond architecture alone, including the number of ciphertext samples and input differences, the sophistication of feature engineering techniques, and the experimental design variations. This complexity underscores the critical need for comprehensive benchmarking studies to establish definitive conclusions about optimal approaches, a challenge we address in detail in subsection X-A.

Unlike natural language processing, neural cryptanalysis has not consistently benefited from the "bigger is better" scaling paradigm described by Kaplan *et al.* [233]. Research has not conclusively demonstrated that deeper or wider neural architectures reliably improve distinguishing capability in cryptographic contexts. Notably, Gohr [17] employed shallower architectures for distinguishers targeting near-uniform ciphertext distributions (specifically for 7 and 8 rounds of SPECK encryption). Differential ciphertext distributions contain subtle non-uniform statistical properties that remain challenging to capture. This underscores a fundamental challenge in developing neural networks capable of effectively learning these cryptographic statistics – a problem requiring sophisticated modeling approaches, which we examine thoroughly in subsection X-B.

b) Multi-Pair Distinguishers ($n > 2$): Neural distinguishers that process multiple ciphertext pairs simultaneously have historically shown minimal practical advantages over simpler approaches [23], [174]. While these complex multi-pair architectures typically performed equivalently or worse than single-pair distinguishers with basic score aggregation, recent evidence suggests this paradigm is shifting – particularly for lightweight block ciphers. Our comprehensive analysis reveals that for SPECK, SIMON, and SIMECK, multi-pair distinguishers have successfully broken more rounds than their single-pair counterparts. This development closely matches common notions in differential cryptanalysis: as the number of rounds increases, the differential probability decreases, and more data is needed to observe a bias; grouping multiple pairs into a sample artificially increases the chance that a rare but relevant differential propagation will occur within each sample.

Interestingly, similar ideas have been widely studied in the machine learning community, in particular under the name of Multiple Instance Learning [234] (MIL), but the corresponding techniques have so far not been applied at all in the context of neural distinguishers. A typical benchmark for MIL is the Elephant dataset, introduced in [235], where the samples are groups of images, with a positive label if the group contains an elephant, and a negative label otherwise. This problem mirrors the case of high rounds neural distinguishers, where most pairs are not helpful, but rare pairs that follow a 'good' differential pattern (the elephants) determine the label. Recent approaches to the MIL problem, such as [236], seem to be promising directions to explore in order to improve multi-pair classifiers. Similarly, the problem of *anomaly detection* has received considerable attention in the machine learning community; if we choose to treat the 'elephant pair' as an anomaly to an otherwise unremarkable distribution, adapting approaches such as Deep One-Class Classification [237] could yield interesting results. Finally, the Deep Set framework [238] considers functions of sets, and addresses issues such as permutation invariance, which are relevant to multiple pair classification, for which the order of the pairs has no importance.

This evolving effectiveness of multi-pair architectures represents a significant development and offers a promising direction for future cryptanalytic research; however, this line of work has so far largely ignored the significant body of work available in the deep learning community, and we believe there is significant room for improvement through incorporating these techniques.

c) Using Multiple Input Differences ($m > 1$): The effectiveness of differential cryptanalysis using multiple input differences has been demonstrated across several cipher families, including SIMON, SIMECK, and SPECK. Gohr *et al.* [23] established a crucial relationship: neural differential distinguisher accuracy correlates with the statistical distance between the separated ciphertext-difference distributions (particularly when distinguishing from a uniform distribution $E = R$). Distinguishers naturally perform better when targeting input differences that produce more distinguishable output distributions.

Despite this advantage, integrating multiple-difference approaches into practical key recovery attacks presents significant challenges. The current attack framework pioneered by Gohr [17] fundamentally relies on distinguishing real ciphertext distributions from uniform distributions as its core mechanism. Adapting this framework to leverage the statistical power of multiple input differences would require substantial modifications to the underlying cryptanalytic methodology.

One promising research direction is exploiting structural relationships between differential characteristics through switching bits for adjoining differentials (SBfADs) [166]. For multi difference distinguishers, requirements could be relaxed to conformance with any one output difference, rather than requiring all differentials to share identical output differences.

d) Feature Engineering (T): Feature engineering has demonstrated a significant impact on distinguisher performance [192], with notable examples including partial decryption and combining ciphertext values with difference-related features. Interestingly, virtually all multi-pair distinguishers obtaining state-of-the-art results utilize advanced feature engineering. Nevertheless, well-designed network architectures can autonomously learn optimal feature representations, as demonstrated by Gohr *et al.* [23] for SIMON, which achieved comparable results to the feature engineering of Bao *et al.* [166].

The explainability analyses in [165] demonstrate that Gohr's architecture $\mathcal{N}_{\text{Gohr}}$ effectively utilizes information beyond mere ciphertext differences ($C_1 \oplus C_2$). Consequently, architectures incapable of efficiently extracting XOR differences from ciphertexts and feature engineering limited to solely ciphertext differences ($F = \delta$) exhibit fundamental constraints, failing to capture the full spectrum of cryptanalytically relevant features.

The extent to which hand-crafted features enhance neural network learning capabilities remains an open research question. Establishing comprehensive benchmarking frameworks would provide valuable insights into the relative merits of automated versus engineered feature extraction for cryptanalytic applications (subsection X-A).

e) Alternative Adversarial Models (E): Paralleling classical cryptanalysis, adversarial models with expanded capabilities consistently outperform against increased cipher rounds, as demonstrated by related-key and conditional approaches extending several rounds beyond chosen plaintext counterparts. Rotational cryptanalysis and other specialized techniques have also shown promising results when adapted to neural frameworks, exploiting structural weaknesses that conventional differential approaches might miss.

The critical research question is what additional adversarial models remain unexplored. Classical cryptanalysis offers numerous attack vectors yet to be fully adapted to neural network distinguishers. Systematically mapping these techniques to their neural counterparts could reveal new attack classes.

IX. Neural Distinguisher Training: Best Practices

Neural network training is not a deterministic process: it is subject to significant variations in the outcome that are caused, for example, by the (random) network parameter initialization process, and the batch process of training data and corresponding differing movement through the optimization plane. Further, the chosen hyperparameters and neural network architectures heavily influence the training outcome.

To interpret the success of neural network training correctly, it is important to distinguish between training, validation, and test data carefully. Each dataset has an important role: The *training data* is used to calculate the loss of the model and to update the model parameters. However, the goal of neural network training is not good performance (low loss) on known data, but instead, generalization to previously unseen data. To monitor the model's performance on previously unseen data during training, *validation data* is used.

A commonly observed phenomenon during neural network training is *overfitting*. At some point during the training, the model does not learn new generalizable features of the training data but instead uses its parameters to learn the training dataset "by heart". This leads to an increasing validation loss. Instead of using the model that has been trained for the maximum number of epochs, in this case, one better uses the model with the minimum *validation data loss*. However, the validation data has now been used in model optimization and can no longer be used to characterize performance based on previously unseen data. Fresh *test data* should be used for the final characterization instead.

The number of parameters of a deep neural network *does not* relate to its computational training cost straightforwardly. Instead, it depends on the computations required by the particular layers used in the network model. The computational training cost should be measured in terms of the required number of FLOPs (floating point operations) or MACs (multiply-accumulate operations). Popular deep learning libraries such as TensorFlow and PyTorch provide routines to obtain neural network parameter counts as well as FLOPs.[6] For example, FLOPs can be evaluated with the TensorFlow Keras module `keras-flops`, and the TensorFlow native routine `model.count_params()` provides the parameter count.

Commonly Overlooked Best Practices for Neural Distinguisher Training

1) **Results Reporting I:** Clearly indicate the results obtained on training, validation, and test datasets and the size of each dataset.
2) **Results Reporting II:** Denote accuracy (or any other metrics) with error margins on multiple sets of *freshly generated* test data.
3) **Neural Network Reporting:** Indicate the network's memory requirements using FLOPs and the number of neural network parameters, and training time per epoch on the specific computational environment (e.g., number and type of GPUs or CPUs).
4) **Open Reproducibility:** Publish the code and trained model parameters to enable review, replication, and future comparisons.

Though not unique to neural differential cryptanalysis, these best practices were frequently overlooked in papers during our literature review, underscoring the importance of emphasizing these standards.

X. Future Challenges

A. The Benchmarking Challenge

As the field of neural cryptanalysis grows, it is becoming more difficult to compare different works on a given primitive due to significant variability in the architectures used, training regimes, distinguishing experiments, or feature engineering. To gain a better understanding of neural distinguisher, we see the creation of a benchmarking platform as an important challenge in the medium term. The goal of such a platform would be to compare neural architectures submitted by authors on sets of standard problems and compare them in a leaderboard. This objective is, however, not straightforward, and we discuss some friction points below.

a) Defining Problems: A problem can be defined as an n-M-T-E configuration, a primitive, a training pipeline, and a dataset size. A logical first step would be evaluating all models on the initial SPECK32 problem in the 2-1-CT-R setting to identify top-performing architectures.

Training regimes are critical: Gohr's work [17] required an advanced pipeline with pre-training on likely differences followed by re-training with $100\times$ more samples to reach 8 rounds. Subsequent research often employs similar polishing techniques. This creates a distinction between raw performance (training from scratch under consistent conditions) and enhanced accuracy (using pretraining [232], layer freezing [23], previous-round distinguisher retraining [20], or increased final-round samples). A standardized pipeline for comparing enhanced distinguishers would be beneficial.

Sample quantity also matters. Many works follow Gohr's approach [17] (10^7 training, 10^6 test samples), as reduction significantly impacts performance. Multiple-pair sample approaches [44] present comparison challenges: fixing sample count gives unfair advantages to models seeing more pairs, while fixing pair count may disadvantage models trained on fewer samples (extreme case: 10^7 pairs per sample would mean training on a single sample). Despite some works using over 1 billion ciphertexts, little research explores this data magnitude in the 2-1-CT-R scenario versus multiple-pair approaches – an axis worth including in benchmarking studies.

b) Metrics: The first challenge to comparing different models is to define what is to be compared. As of now, the main metrics used to compare neural distinguishers are accuracy, true positive rate, true negative rate, and more recently [42], the number of floating point operations (FLOPS), which impacts the training time and quantifies the time complexity of the inference part in a key recovery attack. In the Deep Learning community, the EfficientNet framework [239], which proposes techniques to scale a neural network based on inference speed or parameter count constraints, is often used as a baseline comparison for new models. For neural distinguishers, we could similarly use the number of parameters and FLOPs ratio with the original architecture from Gohr, providing context to the obtained accuracy. However, we also need dedicated metrics adapted to the specific use cases of neural cryptanalysis. In particular, the current metrics do not provide much information on the key recovery complexity, which largely depends on the *wrong key response profile* (see subsection IV-C), prepended differentials, and neutral bits.

B. The AI-$\mathcal{ND}$ Challenge

The neural network architectures currently employed in Neural Differential Cryptanalysis have origins that trace back several years. For instance, the Inception Module by Google researchers was introduced in a seminal paper in 2014 [241]. Similarly, Kaiming He *et al.* [242] won the ILSVRC (ImageNet Large Scale Visual Recognition Challenge) 2015 using ResNet. Attention was introduced in "Attention is all you

[6]The performance of libraries for training neural distinguishers has been compared in [168].

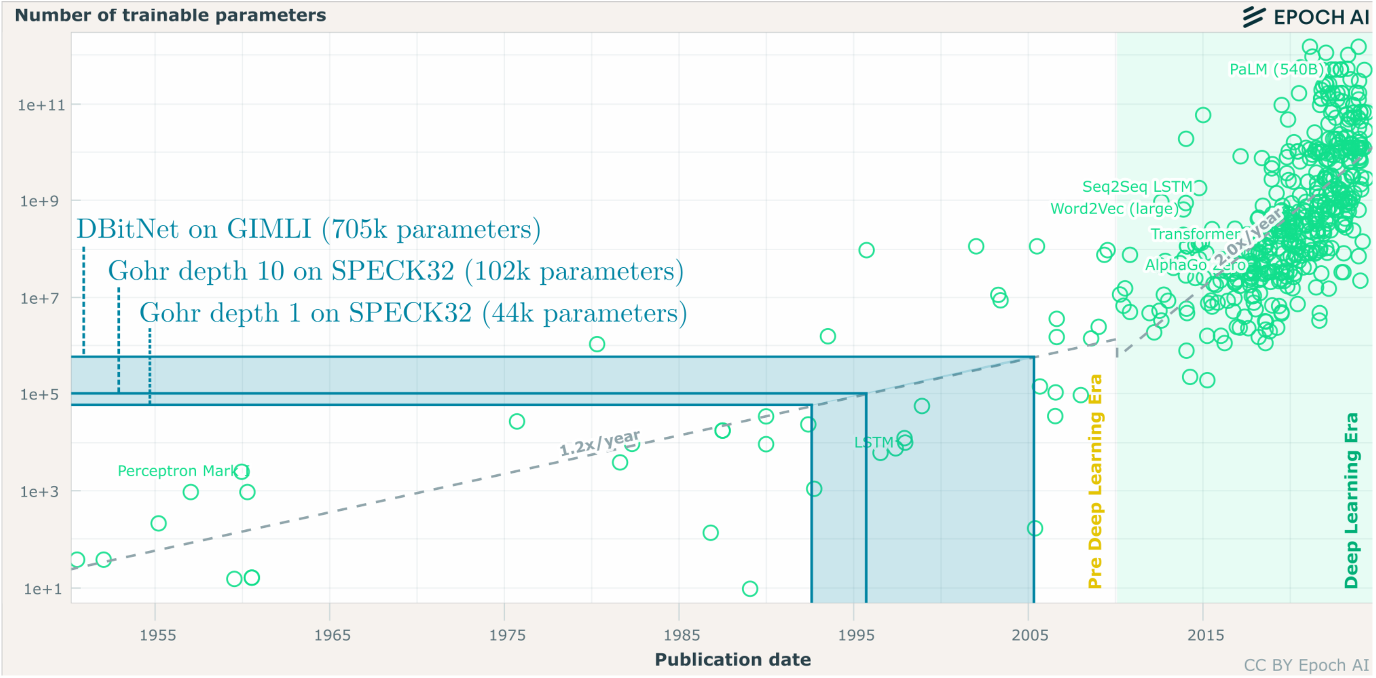

Fig. 4. Adapted from [240] with added data for Gohr's $\mathcal{ND}_{\mathrm{Gohr}}$ on Table III, and DBitNet on Table XI from [42, Table 5].

need" at NeurIPS 2017 [8], and Squeeze-and-Excitation Networks at CVPR 2018 [243].

In recent years, deeper and more complex models have led to a larger parameter count. Figure 4 illustrates the general trend of the increasing parameter count in deep learning models. This is particularly evident in the case of Large Language Models (LLMs), such as GPT, which contain billions of parameters. The deep learning models used to date in Neural Differential Cryptanalysis have low parameter counts compared to more modern "Deep Learning Era" models. Challenges when increasing the model parameter count are higher computational load, longer training times, and overfitting.

However, the advancement of AI technologies such as transformers and reinforcement learning, coupled with increased computational power, holds significant potential for enhancing cryptographic neural differential distinguishers. Transformers, with their capability to handle long-range dependencies and their effectiveness in capturing complex patterns, offer a robust framework for analyzing cryptographic data. Reinforcement learning, on the other hand, provides a powerful approach for optimizing neural network performance through iterative feedback and learning from interactions. These advanced AI methodologies, when applied to cryptographic neural differential distinguishers, can lead to more accurate models. The increased computational power available today allows for training deeper and more complex networks, which can explore a larger hypothesis space and uncover subtle cryptographic weaknesses that simpler models might miss.

Up until now, cryptographers have mainly attempted to apply AI models. As illustrated in subsection X-A, a leaderboard with cryptographically meaningful metrics should be established. Based on the existence of transparent metrics, the **AI-$\mathcal{ND}$ Challenge** aims at (i) motivating cryptographers to use more advanced AI technologies, but also at (ii) motivating cryptographers to establish an AI-competition[7] to allow AI researchers and engineers to apply state-of-the-art methods to Neural Differential Cryptanalysis.

XI. Conclusions

In this paper, we perform a systematic review of the follow-ups to Gohr's seminal paper on neural distinguishers. In the process, we identify and classify works focusing on training neural distinguishers. This systematic review uncovered a young yet vast body of research and a need for common methodological guidelines to grow the field, which we attempt to provide. We also identified two challenges, namely comparing neural distinguisher results and scaling up to much larger and more ambitious architectures.

Over the past 6 years, multiple new settings have been explored for differential cryptanalysis, using multiple pairs per sample or polytopic differences, with the same or varied keys across samples. In addition, various types of feature engineering, particularly through partial inversion, have been explored. These address the question of what clues we can give the neural distinguisher, and multiple avenues are left to explore in that direction. But more fundamentally, what matters perhaps more is what question we ask the neural distinguisher, given this clue, or said differently, what task we ask the neural network to perform. So far, a large portion of the literature has focused on differential-based property for one pair and one input difference, but many variations could be built, as well as tasks related to different types of cryptanalysis or entirely new distinguishing experiments.

[7] Small AI-competitions are hosted on platforms such as Kaggle, while large AI-competitions include the *"Makrikadis"* time series forecasting competition [244], or ILSVRC [245].

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Appendix

A. Comparative Review of the remaining Neural Differential Distinguishers

1) AES: AES is a widely used block cipher standardized by NIST in 2001, designed for general-purpose encryption

TABLE IV
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR AES.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
AES-128	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-$\mathrm{CT}_{\mathrm{tr}}$-R	20M	2M	-	2	0.9981	[225]
	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R	20M	2M	-	2	1	[225]

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
RK Related key setting.

applications. It operates on 128-bit blocks and supports key sizes of 128, 192, or 256 bits. The cipher's structure comprises 10, 12, or 14 rounds (depending on the key size), each involving four transformations: SubBytes (substitution), ShiftRows (permutation), MixColumns (linear mixing), and AddRoundKey. Notably, AES's SubBytes transformation uses a single 8-bit S-box followed by an affine transformation.

Zhang *et al.* [225] developed neural distinguishers targeting a 2-round reduced version of AES-128, specifically using only one pair of bytes from the ciphertext.

2) ARADI: ARADI is a low-latency block cipher introduced by the NSA in 2024, specifically designed for memory encryption applications. It operates on 128-bit blocks and utilizes a 256-bit key. The cipher's structure comprises 16 rounds, each involving a combination of substitution and permutation operations. Notably, ARADI's round function employs a unique S-box, a linear layer, and a key addition layer.

3) ASCON: ASCON is an SPN-based permutation with an input size of 320 bits. It can be used within a sponge construction to build the authenticated ciphers ASCON-128 and ASCON-128a, both using 128-bit keys and 12 rounds in the initialization, and respectively 64 and 128-bit messages, and 6 and 8 rounds in the encryption process. The hash function ASCON-hash, also based on sponge construction, hashes 64-bit messages over 12 rounds. ASCON was announced as the winner of the NIST Lightweight Cryptography Competition in February 2023.

Shen *et al.* [204] trained neural differential distinguishers for the 4-round ASCON with an accuracy of 0.5069 in the standard setting and were able to improve the accuracy to 0.6925 by training another neural network to classify based on the distribution of multiple scores. We do not include this result in the table, as it is a system where the neural distinguisher part is run separately on single pairs rather than a neural distinguisher accepting multiple pairs.

4) CHASKEY: CHASKEY is an ARX-based permutation with an input size of 128 bits. It operates through 8 rounds of Addition, Rotation, and XOR operations on four 32-bit words

5) DES: DES (Data Encryption Standard) is a 16-round SPN block cipher working with 56-bit keys and 64-bit blocks.

Zhang et al. [223] used a staged training approach to obtain a distinguisher for 7-round DES: $4 \cdot 10^7$ samples, 16 pairs each (640M ciphertext pairs).

6) FF1 and FF3: FF1 and FF3 are format-preserving encryption algorithms, with 10 and 8 rounds, respectively, with block sizes of 32 and 128 bits and key sizes of 128 bits. We use the notations FFX-D when the domain is digits and FFX-L when the domain is lowercase characters.

In [185][†], the authors performed neural cryptanalysis of FF1 and FF3 for digits (FFX-D) and lowercase letters (FFX-L). We report the best results in the 2-1-CT-R setting, but note that the authors additionally performed experiments in the m-2-CT-D setting with similar, yet not directly comparable, results. Experiments were conducted for the classification of up to 15 input differences. However, it is not immediately clear which results are the best. The number of samples for training and testing was not specified, nor is the source code available.

7) GIFT: GIFT is a PRESENT-inspired SPN cipher, using 128-bit keys to encrypt 64-bit (GIFT64) or 128-bit (GIFT128) blocks for 28 and 40 rounds, respectively. GIFT was one of the finalists of the NIST Lightweight Cryptography Competition.

In [220][†], the authors claimed a distinguisher on 7 rounds because the training accuracy was 0.6487, despite the validation accuracy being non-significant (0.5002); in the table, we report this 7 rounds distinguisher as it is the best one claimed by the authors, but also their 6-round distinguisher, which has a significant validation accuracy.

In [195][†], the authors claimed a full round distinguisher on GIFT-64 with over 90% accuracy, using 2^{20} samples in total, of which 15% are kept for validation, respectively testing, and a MLP architecture; they also claimed a full round distinguisher on PRIDE with 100% accuracy. Full-round attacks on modern and reputable ciphers are an extraordinary claim and require extraordinary evidence, which the author's manuscript does not provide.

In [199][†], only 10K samples were used for training and testing; as a result, the distinguishers exhibit significant overfitting (e.g., 92% training accuracy and 25% test accuracy for M1 on 6 rounds).

8) GIMLI: GIMLI is a 24-round permutation acting on 384 bits, from which a hash function GIMLI-HASH and an authenticated cipher GIMLI-CIPHER are derived.

9) GOST: GOST is a block cipher developed by the Soviet Union. It operates on 64-bit blocks with a 256-bit key and follows a Feistel network structure with 32 rounds. Each round applies a key-dependent substitution using fixed S-boxes, followed by modular addition and bitwise rotations to ensure diffusion and security.

10) HIGHT: HIGHT is a 32-round ARX-based block cipher operating on 64-bit blocks and 128-bit keys.

Seok et al. [19] achieved a distinguishing accuracy of 0.5707 on 10-round HEIGHT by analyzing only half of the ciphertext

TABLE V
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR ARADI.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
ARADI	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R	20M	2M	✓	5	0.5954	[123]
ARADI $^{\mathrm{RK}}$	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R	20M	2M	✓	6	0.5631	[123]

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
$^{\mathrm{RK}}$ Related key setting.

TABLE VI
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR ASCON.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
ASCON	MLP	3-2-δ-D	1.1M	1.1M	-	3	0.9861	[22]
	MLP	2-1-δ-R	17M	2M	-	4	0.502	[218]
	MLP	2-1-δ-R	20M	20M	-	4	0.5069	[204]
ASCON$^{\mathrm{Unkeyed}}$	Classical ML	2-2-δ-D	64K	16K	-	3	0.916	[168]‡

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
‡ The use of a small validation set raises concerns about the statistical robustness, reproducibility, and generalizability of the results, as such datasets are prone to high variance and may not reliably reflect model performance.

TABLE VII
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR CHASKEY.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
CHASKEY-PERMUTATION	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R	17M	40K	-	4	0.6161	[22]‡
	$\mathcal{ND}_{\mathrm{Gohr}}$	32-1-CT-R	20M	2M	-	4	0.7712	[174]
	INC	16-1-CT-R	60M	2M	-	5	0.5181	[223]

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
‡ The use of a small validation set raises concerns about the statistical robustness, reproducibility, and generalizability of the results, as such datasets are prone to high variance and may not reliably reflect model performance.

TABLE VIII
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR DES.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
DES	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R	20M	2M	-	5	0.58	[173]
	$\mathcal{ND}_{\mathrm{Gohr}}$	4-1-CT-R	20M	2M	-	6	0.5653	[174]
	INC	32-1-CT-R	1280M	32M	-	7	0.5114	[223]

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).

difference (4 out of 8 bytes). These specific bytes were selected through their analysis of the HIGHT round function.

Bose et al. [172]† claimed advancements in distinguishing additional encryption rounds through sequential model training on ciphertext pairs. However, these findings contradict established cryptographic theory, which shows distinguishability decreases monotonically with increasing rounds; a pattern absent in their results. Notably, Bellini et al. [42] reported superior distinguishers for rounds 9 and 10 of HIGHT, challenging the purported architecture's effectiveness in detecting differential patterns. Given these theoretical and empirical inconsistencies, independent verification is necessary before accepting these anomalous results. We report the best distinguishers with statistical significance at the 95% confidence level (z-scores > 1.96, $p < 0.05$), corresponding to accuracies above 0.5011 on a test set of 5×10^6 samples.

11) KATAN: KATAN is a family of FSR-based block ciphers with block sizes 32, 48, or 64, key size 80, and 254 rounds.

For KATAN32, [42] reached statistically significant accuracies up to 69 rounds in an automatically generated distinguisher, and noted that this can be improved to a 71-round

TABLE IX
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR FF.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
FF1-D	MLP	2-1-CT-R	/	/	-	10	0.855	[185]†
FF1-L	MLP	2-1-CT-R	/	/	-	2	0.522	[185]†
FF3-D	MLP	2-1-CT-R	/	/	-	8	0.977	[185]†
FF3-L	MLP	2-1-CT-R	/	/	-	2	0.554	[185]†

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
† A critical discussion of these results is provided in the text.
/ Unknown quantity.

TABLE X
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR GIFT.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
GIFT-64	UNet	12-1-$\mathcal{A}$-R	/	/	-	4	0.725	[219]†
	LSTM	3-2-CT-R	17M	4M	-	6	0.5754	[21]
	MLP	2-1-δ-R	2.1M	300K	-	FULL	0.96	[195]†
GIFT-128	MLP	2-1-δ-R	17M	2M	-	7	0.55	[218]
	MLP	2-1-δ-R	20M	2M	-	7	0.5542	[204]
TweGIFT-128	MLP	2-1-$\mathcal{CT}$-R	2M	200K	-	6	0.5675	[220]
	MLP	2-1-$\mathcal{CT}$-R	2M	200K	-	7	0.5002	[220]†
GIFT-COFB	MLP	2-4-δ-D	$20K$	$20K$	-	4	0.615	[199]†

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
/ Unknown quantity.
† A critical discussion of these results is provided in the text.

distinguisher with 0.5034 ± 0.0002 accuracy using their simple polishing step. In contrast, [186] reached 51 rounds in the standard setting and 59 when using 64 pairs.

In [186], practical key recoveries were obtained for 125, 106 and 95 rounds respectively, in the related key scenario. Single-key conditional neural distinguishers were also mentioned in [186] for 85, 72 and 61 rounds respectively, but the $r + s$ decomposition was not explicitly mentioned so we omit them in the table.

12) KNOT: KNOT is an SPN-based permutation acting on a 256, 384, or 512-bit state; when used in a MonkeyDuplex construction to build a cipher, it uses 28 to 52 rounds, depending on the version.

13) LBCIoT: LBCIoT is a 32-round block cipher encrypting 32-bit plaintexts with an 80-bit key. In [207], the authors propose a neural distinguisher on 7 rounds and build a practical key recovery attack for 8 rounds.

14) LEA: LEA is an ARX-based block cipher, encrypting 128-bit plaintexts with 128-, 192-, or 256-bit keys for 24, 28, or 32 rounds, respectively.

For LEA, [42] proposes the first neural differential distinguisher, reaching 11 rounds with accuracy 0.5109. In comparison, the proposal of LEA [246] presents a differential characteristic with probability 2^{-98} for 11 rounds, and 2^{-128} for 12 rounds.

Bose et al. [172]† claimed advancements in distinguishing additional encryption rounds through sequential model training on ciphertext pairs. However, these findings contradict established cryptographic theory, which shows distinguishability decreases monotonically with increasing rounds – a pattern absent in their results. Notably, Bellini et al. [42] reported superior distinguishers for rounds 9 and 10 of HIGHT, challenging the purported architecture's effectiveness in detecting differential patterns. Given these theoretical and empirical inconsistencies, independent verification is necessary before accepting these anomalous results. We report the best distinguishers with statistical significance at the 95% confidence level (z-scores > 1.96, $p < 0.05$), corresponding to accuracies above 0.5011 on a test set of 5×10^6 samples.

15) PRESENT: PRESENT is an SPN-based block cipher, encrypting 64-bit blocks with 80 (PRESENT-80) or 128-bit keys (PRESENT-128) for 31 rounds.

In [42], a 9-round distinguisher with an accuracy of 0.5092 was given, which favorably compares to the 7-round distinguishers of [174], despite [174] using four pairs per sample. On the other hand, [73] obtained a slightly higher accuracy

TABLE XI
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR GIMLI.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
GIMLI	DBitNet	2-1-CT-R	20M	2M	✓	11	0.524	[42]
GIMLI-HASH	MLP	3-2-δ-D	400K	40K	-	8	0.5219	[22]‡
GIMLI-CIPHER	MLP	3-2-δ-D	400K	40K	-	8	0.5099	[22]‡

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
‡ The use of a small validation set raises concerns about the statistical robustness, reproducibility, and generalizability of the results, as such datasets are prone to high variance and may not reliably reflect model performance.

TABLE XII
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR GOST.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
GOST	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R	2M	200K	-	9	0.5430	[208]
GOST $^{\mathrm{RK}}$	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R	2M	200K	-	14	0.7134	[208]

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
$^{\mathrm{RK}}$ Related key setting.

TABLE XIII
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR HIGHT.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
HIGHT	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R	2M	200K	-	9	0.7472	[208]
	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-δ_{tr}-R	20M	2M	-	10	0.5707	[19]
	DBitNet	2-1-CT-R	20M	2M	✓	10	0.751	[42]
	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R	2M	200K	-	11	0.7472	[208]
	LSTM	2-1-A-R	10M	500K	-	15	0.5015	[172] †
HIGHT $^{\mathrm{RK}}$	DBitNet	2-1-CT-R	20M	2M	✓	14	0.563	[42]
	DenseNet	2-2-CT-D	20M	2M	✓	14	0.640	[214]

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
$^{\mathrm{RK}}$ Related key setting.
† A critical discussion of these results is provided in the text.

at the cost of using 32 ciphertexts per sample. In comparison, the best differential characteristic for PRESENT reduced to 9 rounds has probability 2^{-36} [247].

Bose et al. [172]† claimed advancements in distinguishing additional encryption rounds through sequential model training on ciphertext pairs. However, these findings contradict established cryptographic theory, which shows distinguishability decreases monotonically with increasing rounds – a pattern absent in their results. Notably, Bellini et al. [42] reported superior distinguishers for rounds 9 and 10 of HIGHT, challenging the purported architecture's effectiveness in detecting differential patterns. Given these theoretical and empirical inconsistencies, independent verification is necessary before accepting these anomalous results. We report the best distinguishers with statistical significance at the 95% confidence level (z-scores > 1.96, $p < 0.05$), corresponding to accuracies above 0.5011 on a test set of 5×10^6 samples.

16) PRIDE: PRIDE is a 20-round SPN cipher using 64-bit blocks and 128-bit keys.
In [195], the authors claimed a full-round distinguisher on the cipher with 100% accuracy, which seems likely to be attributed to a methodology issue rather than an actual break, as perfect accuracy is often a sign of overfitting, especially considering the lack of evidence provided in the paper.

17) SHA3: SHA3-256 is a 24-round sponge-based hash function with an output size of 256.

18) SKINNY: SKINNY is an SPN-based block cipher; SKINNY128 processes 128-bit plaintexts with 128, 256, and 384-bit keys for 40, 48, and 56 rounds, respectively.

19) SLIM: SLIM is a 32-round block cipher encrypting 32-bit plaintexts with an 80-bit key.

TABLE XIV
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR KATAN.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
KATAN32	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-δ-R	20M	2M	-	51	0.533	[186]
	$\mathcal{ND}_{\mathrm{Gohr}}$	128-1-δ-R	1280M	128M	-	59	0.575	[186]
	DBitNet	2-1-CT-R	20M	2M	✓	69	0.505	[42]
KATAN32 $^{\mathrm{C}}$	$\mathcal{ND}_{\mathrm{Gohr}}$	64-1-δ-R	64M	6.4M	-	58	0.602	[189]
	$\mathcal{ND}_{\mathrm{Gohr}}$	128-1-δ-R	1280M	128M	-	85	0.570	[186]
KATAN32 $^{\mathrm{RK,C}}$	$\mathcal{ND}_{\mathrm{Gohr}}$	128-1-δ-R	1280M	128M	-	112	0.647	[186]
KATAN48	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-δ-R	20M	2M	-	40	0.58	[186]
	$\mathcal{ND}_{\mathrm{Gohr}}$	96-1-δ-R	960M	96M	-	50	0.54	[186]
KATAN48 $^{\mathrm{C}}$	$\mathcal{ND}_{\mathrm{Gohr}}$	64-1-δ-R	64M	6.4M	-	47	0.582	[189]
	$\mathcal{ND}_{\mathrm{Gohr}}$	96-1-δ-R	960M	96M	-	72	0.582	[186]
KATAN48 $^{\mathrm{RK,C}}$	$\mathcal{ND}_{\mathrm{Gohr}}$	48-1-δ-R	960M	96M	-	96	0.625	[186]
KATAN64	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-δ-R	20M	2M	-	31	0.718	[186]
	$\mathcal{ND}_{\mathrm{Gohr}}$	128-1-δ-R	1280M	128M	-	36	0.548	[186]
KATAN64 $^{\mathrm{C}}$	$\mathcal{ND}_{\mathrm{Gohr}}$	64-1-δ-R	64M	6.4M	-	26	0.613	[189]
	$\mathcal{ND}_{\mathrm{Gohr}}$	128-1-δ-R	1280M	128M	-	61	0.613	[186]
KATAN64 $^{\mathrm{RK,C}}$	$\mathcal{ND}_{\mathrm{Gohr}}$	128-1-δ-R	1280M	128M	-	86	0.728	[186]

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
$^{\mathrm{RK}}$ Related key setting.
$^{\mathrm{C}}$ Conditional setting.

TABLE XV
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR KNOT.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
KNOT-256	MLP	3-2-δ-D	1.6M	1.6M	-	10	0.5912	[22]
KNOT-512	MLP	3-2-δ-D	1.6M	1.6M	-	12	0.6032	[22]

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).

TABLE XVI
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR LBCIOT.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
LBC-IoT	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R	2M	200K	-	7	0.6408	[207]‡

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
‡ The use of a small validation set raises concerns about the statistical robustness, reproducibility, and generalizability of the results, as such datasets are prone to high variance and may not reliably reflect model performance.

In [200], the authors performed experiments with low key entropy (10 and 100 keys, respectively, for 1M samples), as well as with one random key per sample. We report the last one for comparability and note that the results were very close in the 3 cases.

In [207]†, the reported accuracy is 0.5036 on 10^5 samples, which corresponds to less than 3 standard deviations and has a probability over 1% of occurring for distinguisher making predictions at random; we question the relevance of this result, as testing on more data is required to prove statistical significance.

20) TEA and XTEA: TEA and its successor XTEA are 64-round block ciphers encrypting 64-bit plaintexts with a 128-bit key.

In [20], the authors considered modular addition-based differentials, where the input difference is injected by modular addition, which we denote by R^+ in the experiment. [42] automatically found distinguishers for both TEA and XTEA for 5 cycles (10 rounds), respectively, with accuracies 0.5634 and 0.5984; the authors noted that they interestingly share the same input difference. For TEA, [42] reached two more rounds than [20].

TABLE XVII
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR LEA.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
LEA-128	DBitNet	2-1-CT-R	20M	2M	✓	11	0.512	[42]
	Transformer	2-1-A-R	10M	500K	-	13	0.5012	[172]†

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).

TABLE XVIII
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR PRESENT.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
PRESENT-64/80	$\mathcal{ND}_{\mathrm{Gohr}}$	8-1-CT-R	20M	2M	-	7	0.5853	[174]
	UNet	12-1-$\mathcal{A}$-R	/	/	-	7	0.664	[219]‡
	DBitNet	2-1-CT-R	20M	2M	✓	8	0.512	[42]
	$\mathcal{ND}_{\mathrm{Gohr}}$	2-2-δ-D	20M	2M	-	8	0.515	[210]
	INC	32-1-CT-R	960M	32M	-	8	0.5416	[223]
	LSTM	2-1-A-R	10M	500K	-	12	0.5014	[172] †
PRESENT-64/80 $^{\mathrm{RK}}$	MLP	6-1-δ-R	4.2M*	1.9M*	-	5	0.614	[197]
	$\mathcal{ND}_{\mathrm{Gohr}}$	2-2-δ-D	20M	2M	-	10	0.517	[210]

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
$^{\mathrm{RK}}$ Related key setting.
‡ The use of a small validation set raises concerns about the statistical robustness, reproducibility, and generalizability of the results, as such datasets are prone to high variance and may not reliably reflect model performance.

TABLE XIX
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR PRIDE.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
PRIDE	MLP	2-1-δ-R	2.1M	300K	-	20	1	[195]‡

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
‡ The use of a small validation set raises concerns about the statistical robustness, reproducibility, and generalizability of the results, as such datasets are prone to high variance and may not reliably reflect model performance.

21) TinyJAMBU: TinyJambu-128 is an authenticated encryption algorithm based on a 640-round NLFSR-based permutation, which encrypts 128-bit blocks. TinyJambu-128 was among the ten NIST's lightweight cryptography finalists.

In [21]†, the authors claimed a full-round distinguisher on TinyJambu, which we challenge. In the provided code, the ciphertexts in a sample use the same key, nonce, and associated data, which would provide a trivial distinguisher. As noted by the designers of TinyJambu[8]: 'When nonce is reused, an attacker can decrypt the ciphertext since the encryption of TinyJAMBU is somehow similar to the Cipher Feedback mode.'.

[8]https://csrc.nist.gov/CSRC/media/Projects/lightweight-cryptography/documents/finalist-round/updated-spec-doc/tinyjambu-spec-final.pdf

TABLE XX
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR SHA3.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
SHA3-256	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R	2M	2M	-	3	0.9904	[174]

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).

TABLE XXI
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR SKINNY.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
SKINNY128$^{\mathrm{Unkeyed}}$	Classical ML	2-2-δ-D	32K	32K	-	6	0.9912	[168]‡
	Classical ML	2-2-δ-D	2M	2M	-	7	0.5456	[168]

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
‡ The use of a small validation set raises concerns about the statistical robustness, reproducibility, and generalizability of the results, as such datasets are prone to high variance and may not reliably reflect model performance.

TABLE XXII
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR SLIM.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
SLIM	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R	2M	200K	-	3	0.5036	[207]†
	$\mathcal{ND}_{\mathrm{Gohr}}$	2-1-CT-R	2M	/	-	5	0.814	[200]

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
/ Unknown quantity.
† A critical discussion of these results is provided in the text.

TABLE XXIII
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR TEA AND XTEA.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
TEA	MLP	2-1-CT-R^{+}	2M	20K	-	8	0.545	[20]‡
	DBitNet	2-1-CT-R	20M	2M	✓	10	0.563	[42]
XTEA	DBitNet	2-1-CT-R	20M	2M	✓	10	0.598	[42]

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
‡ The use of a small validation set raises concerns about the statistical robustness, reproducibility, and generalizability of the results, as such datasets are prone to high variance and may not reliably reflect model performance.

TABLE XXIV
OVERVIEW OF THE NEURAL DIFFERENTIAL DISTINGUISHERS FOR TINYJAMBU.

Primitive	Arch.	Class	Trn.	Val.	AutoND	Rounds	Acc.	Ref.
TinyJAMBU-128	MLP	2-1-δ-R	2.097M	262K	-	FULL	0.9958	[21]†

Class: n-m-T-E, from subsection VIII-B. Under this convention, Gohr's initial experiments are 2-1-CT-R, and the results obtained in greyed out settings n-m-T-E are not directly comparable. **AutoND:** indicates if the neural distinguisher was automatically generated (✓) or is the result of an elaborate, manually designed training procedure (-).
† A critical discussion of these results is provided in the text.